

WOOD WANDERINGS



WINTHROP PACKARD

WOOD WANDERINGS



WINTHROP PACKARD

The Project Gutenberg eBook of Wood Wanderings

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Wood Wanderings

Author: Winthrop Packard

Illustrator: Charles Copeland

Release date: August 14, 2021 [eBook #66059]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/66059

Credits: Steve Mattern, Chuck Greif and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This book was produced from images made available by the HathiTrust Digital Library.)

*** START OF THE PROJECT GUTENBERG EBOOK WOOD WANDERINGS

[Contents.](#)

[Index.](#)

[List of Illustrations](#)

(In certain versions of this etext [in certain browsers] clicking on the image will bring up a larger version.)

(etext transcriber's note)

WOOD WANDERINGS

THE WORKS OF WINTHROP PACKARD

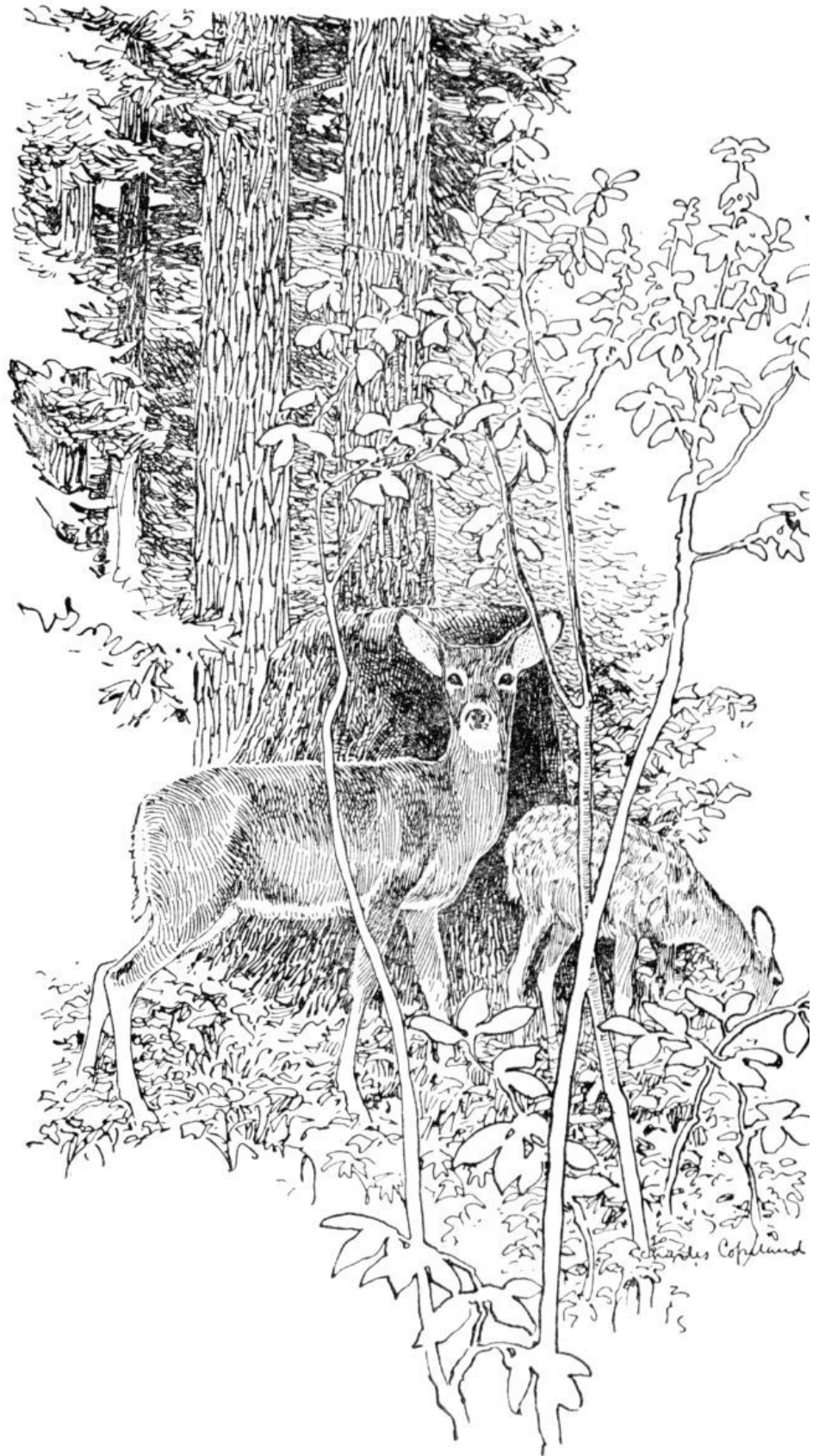
WOODLAND PATHS
WILD PASTURES
WOOD WANDERINGS
WILDWOOD WAYS

Each illustrated by Charles Copeland

12mo. Ornamental cloth, gilt top, each volume \$1.20 *net*, postage
8 cents

The four volumes together constitute "The New England Year,"
dealing, in the order given, with the four seasons. The set, boxed,
\$4.80; *carriage extra*. Sold separately.

SMALL, MAYNARD AND COMPANY
PUBLISHERS BOSTON



You may see a slender doe pirouette like a ballet-dancing wood
nymph
[Page 38]

WOOD WANDERINGS

BY
WINTHROP PACKARD

ILLUSTRATED BY
CHARLES COPELAND



BOSTON
SMALL, MAYNARD AND COMPANY
PUBLISHERS

COPYRIGHT, 1910

BY SMALL, MAYNARD AND COMPANY
(INCORPORATED)

Entered at Stationers' Hall

THE UNIVERSITY PRESS, CAMBRIDGE, U.S.A.

THE author wishes to express his thanks to the "Boston Transcript" for permission to reprint in this volume matter which was originally contributed to its columns.

CONTENTS

	PAGE
FAIRY FRUIT	1
THE LAND OF SPRUCE	21
BIRDS OF THE NOR'EASTER	43
THE SQUIRREL HARVEST	65
AMONG AUTUMN LEAVES	85
THE DAY THAT SUMMER CAME BACK	107
WHEN AUTUMN PASSES	129
NOVEMBER WOODS	149
WINTER BIRDS'-NESTING	171
SOME CROWS I HAVE KNOWN	193
INDEX	217

ILLUSTRATIONS

You may see a slender doe pirouette like a ballet-dancing wood nymph	<i>Frontispiece</i>
	OPPOSITE PAGE
The woodchuck is the very mark and origin of the paunchy gnome	8
Seems to think himself secure there	36
The red squirrel gets the burs after the fashion of the real sport	68
He does not have to look for his food	160
A field mouse had appropriated this nest for an autumn storehouse	182
Across the angry crimson of the west flitted silhouetted black wings	198

FAIRY FRUIT

TO-DAY the September west winds have begun the fall house-cleaning by sweeping the tops of the pine woods. All the morning the little brown scales which nestle close to the base of each pine leaf as it grows, protecting it from the withering force of the midsummer sun, have been soaring and spinning in high glee, curiously lighting up with brown glimmers the solemn sanctuaries beneath.

It is the first prophecy of winter under the sheltering boughs where still lingers the midsummer warmth. The chickadees, going their forenoon rounds, scold about it in a brisk fashion that is in tune with the briskness of the wind itself. In the languor of the south wind the chickadee has a little lazy song which he sings often, "Sleepie, sleepee," a tuneful little ditty that makes you want to stretch out on the brown carpet with a mound of green moss for a pillow and let the resinous odors lull you to sleep. I always feel that the bird himself murmurs it with one eye closed and himself in danger of falling off the perch in slumber.

None of that song to-day. It's "chick-chickachick, chick-a-chicadee dee dee," with a snap in it like the crack of a whip. Yet the flock soon passes on, and in the dreamy warmth of the grove you know little of the vivid touch in the wind. Only enough of it comes through to set the little brown pine motes to whirling merrily as they fall, vanishing from sight like flitting elves as they touch the brown carpet below.

There was another elf-like transformation, an appearing and a disappearing, in the woods this morning. That was a *Pyrameis atalanta* that kept vanishing into the trunk of a big pitch pine. This, the red admiral, own cousin to the familiar *Pyrameis cardui*, the painted lady, is a butterfly whose movements are as snappy as those of the west wind on these house-cleaning days. Rich red, white and black are the colors on the upper side of its wings, but when these are closed there is exposed only the under side, which makes the creature so exactly like a rough chip of the pitch-pine bark that when he lights on the trunk the vanishing is complete. Out of nothing he sprang, a vivid flash of darting red and white flipping before your eyes, then he darted up to the pine trunk that seemed to open and let him go in, so completely did he transform his bright colors into a bit of brown bark.

The more I see of woodland glades and sun-dappled depths and the creatures that inhabit them the less I am inclined to smile at the elder races of the world that peopled them with fairies, sprites, and goblins. Why should they not believe in these things? It is hard sometimes for us to forego all lingering remnants of faith in such inhabitants of field and wood.

This morning on my way to the grove I seemed to meet with more than the usual number of woodchucks, though you would hardly call it meeting, for our paths never crossed. But in three different parts of the big mowing-field a woodchuck bobbed out of nowhere in particular. No doubt he was feeding on the clover of the farmer's

aftermath, but I saw no more of that than the cropped herbage after the woodchuck was gone. My first sight each time was when the animal began to roll in a straight line across the field. I say roll, for woodchucks at this time of year are so fat that they do not seem to run, but undulate over the grass as does the deep sea wave over the shallows.

I never can help chasing them, though I know well what is about to happen. Nor do I expect to catch one, for, fat as they are, they move with surprising rapidity. Even if I happen to know where his hole is by the pile of dirt at the door and rush between him and it, I am no nearer getting my game. I always fancy that the fat shoulders of the woodchuck jiggle with laughter and his little pig eyes twinkle, for that is just what he expects and is prepared for. He keeps right on in his straight line, then psst! he vanishes. You don't see him dive or turn or hide. He just goes out of sight. You may poke about in the grass for a long time before you find the secret entrance by which he has returned to his burrow. Sometimes he has two of them. They are dug from within outward and no tell-tale trace of dirt is left to mark their location. This has all been carried down with infinite pains, then up, and left at the public door, where all may see it. The woodchuck is the very mark and origin of the paunchy gnome, which is said to guard buried treasures, and which bobs out of the earth, frightens Hob from his intended mining, then bobs back into the earth to guard the gold.

So you have but to go into the pine grove to-day with inquiring eye and



The woodchuck is the very mark and origin of the paunchy
gnome

acquiescent mind and all the beautiful old superstitions that always plead to be taken into the belief will come trooping along, to your supreme delectation. Well might the great and good Wordsworth say, he who knew the open wold and the bosky dell as few of us are privileged to know them, and wrote about them as none of us can:

“Great God! I’d rather be
A Pagan suckled in a creed outworn,
So might I, standing on this pleasant lea,
Have glimpses that would make me less forlorn;
Have sight of Proteus rising from the sea,
Or hear old Triton blow his wreathed horn.”

Here in the pine grove is the riding-school of sylphs innumerable,—those fragile fairies who float in slender grace on the passing breeze. Their launching stands are the flat-topped receptacles of the blooms of *Erechthites hieracifolium*, the coarse and homely fireweed. All summer it has stood in the open spaces of the wood with its tall stalks bearing blossoms that look like green druggists’ pestles, with no beauty of petal or sepal to entice, no fragrance to call the wandering bee. Indeed, these surly blooms seem like buds that were too cross to open. Now it is different. The green bonds of guardian bracts are reflexed, and you may now see that this unattractive flower has held close pressed within its homely heart companies of sylphs.

White and slender and soft, they stand until the right wind comes along, then they spring fearlessly to his invisible shoulders and are borne whither they list. Not mortal things are these thistledown fairies that are so transparent white that you may look through them as they float by and see the sun. If it pleases them to touch your hand or your cheek as they pass, you may note an ethereality of sensation which is thought rather than feeling, so light it is.

The *Epilobium angustifolium*, sometimes called willow herb, is another fireweed, as beautiful of bloom as *Erechthites* is homely. Like this, it grows in waste places in the wood, flaunting its long raceme of showy, pink-purple flowers all summer. Like the *Erechthites*, too, when September has tamed its exuberance, it is more beautiful still as the abode of white sylphs which cling in whorls to its stem. Yet, mark you the difference. The sylphs, reared by the dour and homely fireweed, stand erect and prim in close communion as stately and correct and dignified as sylphs may be. Those born of the flaunting *Epilobium* cling to it in graceful, almost voluptuous abandon, assuming such poses as nymphs might in wooing a satyr. Equally beautiful, the first are like prim New England schoolmarms diaphanously gowned for a Greek play; the second suggest artists’ models frolicking in the woodland before being called to pose.

Along with these two fireweeds, breeders of sylphs, in my pine wood grows the pokeweed, a villainous name for a wonderfully vigorous and beautiful plant. Just now its close-set racemes of purple-black berries are ripening, their color a vivid contrast

with the smooth rich green of its ovate-oblong leaves and the wine color of its stems. It is really a royal plant, and so great is its vigor that its dark berries threaten to burst their skins and scatter their rich crimson lifeblood. If you will look closely at the berries you will see that the fairies have stitched them neatly across the top to prevent this. The marks of the needle show, and the tiny puckering made by drawing the thread very tight.

It is so workmanlike a performance that I suspect the *leprachauns*, who are shoemakers, of having been called in to do it,—called in, for the *leprachauns*, without doubt, have all they can do conveniently, making and mending the fairy shoon. No doubt the brownies, who are domestic fairies and who would be keeping watch of the woodland fruits anent the preserving season, had them attend to this, lest the preserving be a failure. The poke berries look so rich and luscious that I have tried them; but I cannot say that I like the flavor, which is rich indeed, but peculiar. But then, I remember my first olive. They don't taste half so bad as that did, and compared with pickled limes, which school-girls eat with avidity, they are nectar and ambrosia in one package.

All the under-pine world is spread just now with beautiful berries, for which neither we nor the birds seem to have a taste. There are the partridge berries, which, by the way, I have never seen a partridge eat, nor have I found them in the crops of partridges, which I have been mean enough to shoot. Yet these are, to my mind, the most edible of all, though they are insipidly sweet, and their flavor is so finely pleasant that it is not for the coarse palate of most mortals. Their vines carpet the wood in places, and the soft, pure red of the berries would catch the eye of bird or beast from afar. These stay ripe and sound all winter, and you may see their red shining softly among the evergreen leaves when the bare ground responds, dull and sleepy still, to the resurrection trump of spring. They have not been gobbled whole, therefore the larger animals and birds of the wood do not care for them; but in the spring you will often find them with a tiny bite taken out of one side. This can have been done by no other than the fairy urchins, too young to eat fruit with safety, and forbidden by their mothers, they yet slip out and take a bite before they can be hindered.

Equally beautiful and conspicuous, and equally insipid to the human taste, are the great blue berries of the *Clintonia borealis*, which grows sparingly under the pines hereabouts. These are as large as the end of your finger, and a wonderful clear shade of prussian blue. If you know the leaf of the lady's slipper,—the moccasin-flowered orchid which is so common in June under all pines,—you might, thinking of the leaf only, call this the fruit of the lady's slipper, where, as sometimes happens, but one berry grows on a stem. Yet if you look further you will not long labor under the mistake, for you will find many stalks with several berries, whereas the single blossom of the *Cypripedium acaule* could leave behind it but one. The fruit of the lady's slipper is at this time of the year a dry brown pod, whence all the little dry seeds have long ago dropped; indeed, it is only occasionally that you will find the pod left so long.

I do not know but birds eat the beautiful fruit of the *Clintonia*, though I have never seen them do it, and I fancy it is too insipid to creatures that love wild blackberries, raspberries, and cherries. Yet, as in the case of the partridge berries, I have often seen the fruit with a tiny mouthful taken out of it as it stands on the stalk. This is a bigger mouthful than the marks left in the partridge berries, so I know that it is not fairy urchins which have done it, even if I thought they could climb these tall, slippery stalks. I have a fancy that Queen Mab herself, who, as you very well know, is the fairy midwife as well as queen, flitting home in the dusk of morning from motherly service, has stopped for a brief refreshment on the *Clintonia* stalk. I even have a notion that I can see in the bitten berries the prints of the wee pearls that are her teeth.

Every little starry bloom of the *Smilacina bifolia*, which vies with the *Mitchella* in carpeting the pine wood, leaves behind it a lovely tiny berry that is like a pinhead currant. These, now, are in little groups at the top of the withering stalks. Fairy currants I have heard them called, and I think the name a good one, for they are red and juicy like currants and taste not unlike them, though, like all these fruits, the flavoring is more insipid. They are a lovelier berry before ripening than after, for when young they are a slender sage green, through which the red shows more and more in dappling spots as they ripen, making them a most beautiful warm gray.

I am quite sure that the fairies make jam of these, stowing it away in wild-cherry stone jars, built for them by the stone-mason wood mice, who are very busy with the wild-cherry stones about this time. They drill a little round hole in each and extract the kernel, then put the stones away in their storehouses for sale to the fairies. I have often found these storehouses with the stones put away in them, but have never been fortunate enough to find the fairy larder with the jam in the jars.

I often wonder what the fairies think of the fruit of the nodding *trillium*, which you will find in the wood now with the others. I fancy they look upon it with wonder and amazement as a miracle of agriculture, just as we, about this time, wonder at the vast pumpkin exhibited at the county fair. It is sometimes almost an inch in diameter, roundish, with six angles or flutings on it, and a very vivid crimson in color.

To the fairies they must seem to grow, like cocoanuts, on palm trees, for the *trillium*'s erect stem, bearing its spreading palm-like leaves only at the top, is a foot or so high. I imagine they gather these as they fall with great glee, and stow them away for winter use in making fairy pumpkin pies. Often in autumn, along woodland paths in the night, I have seen a faint glow where I was about to set my foot. Always I step aside carefully, for I have been told that this soft, greenish light comes from glowworms.

Yet it is more than likely that sometimes the fairy urchins have been allowed to make jack-o'-lanterns from the smaller of these *trillium* pumpkins, and this faint glow is the fairy candle within these. After stepping aside you should bend your head and listen. If you hear faint, tinkling laughter, inexpressibly sweet and fine, it is the urchins

out with their jack-o'-lanterns, and laughing in glee that they have succeeded in scaring someone.

THE LAND OF SPRUCE

THE seamed and wrinkled face of Katahdin, brown and weather-beaten, looks over twenty-five miles of unbroken forest eastward to "Number One" plantation, through which runs the fine gray line of the Patten road. Southward for miles upon miles, northward for miles upon other miles it stretches, taut and straight as a bowstring, narrow as a creed, and as inexorable.

On either side of it, here and there, the hand of man has hewn an open space for a farm. Yet you may stand on the summit of the ridge at Number One and look eastward for forty miles and see only the unbroken green of the forest, with the black lances of the firs and spruce stabbing the sky. The thin gray road seems about to be crushed and wiped off the world by these green eastern and western millstones which press upon it. They smooth off the boundaries of the farm spaces, roll over fences, and crush them into the black earth beneath. The lone farmer fights valiantly against this, but sooner or later old age gets him, or a fire burns his buildings; then the forest rolls majestically on and over him.

That is what it has done up on Number One. On the long white line of the Patten road a single house and farm buildings remain. These mark General Winfield Scott's farthest north during the Aroostook war, three-quarters of a century ago, when Maine and New Brunswick quarreled over boundary lines. I can but fancy that the general, who had traveled that long, thin line of straight road, from Bangor to Lincoln, to Mattawamkeag, and thence to Number One, up hill and down dale, with never a curve to rest the eye or avoid a hill, sighed thankfully when he learned that he need not reach his journey's end.

Along this road in his day, and for fifty years after, trailed the tote teams laden with goods for northern Aroostook, returning weighted with the products of the forest. Four and six-horse teams they were, and they traveled sometimes a dozen in a procession, doubling hitches at some steep pitch and hauling the wagons over, one by one. The road was a busy one then, and the old taverns strung along at intervals of a dozen miles or so rang with life. To-day those that remain are bleak and deserted, and only a few remain. The others have been burned at one time or another.

Along this road came Thoreau on his trip into the Maine woods, and you may yet see the doorstone on which he stood and looked across to the store across the street, which was so diminutive that the stout proprietor, as he said, had to come out to let a customer in. Thoreau might well have been surprised could he have known the volume of business done in this diminutive store, which was really only the office of the big barn behind, which held the goods in bulk. No wonder a proprietor waxes fat when people hitch up and drive fifteen or twenty miles to trade at his store, the only one within that distance.

To-day of South Moluncus not much more than the thresholds remains, the whole village having been wiped out by fire. But the glory of the place had departed long since. The railroad which brings civilization and prosperity to some places takes it away from others; and Mattawamkeag and Kingman thrive, while South Moluncus and other once busy little centers in the virgin forest along the old Patten road are like the cities of old Greece, but memories and ash heaps. The porcupine noses unmolested in many a cellar along the narrow way, the deer browse undisturbed on the apple trees, and over the once prosperous farms passes the resistless, majestic march of the forest.

It cannot subdue that thin gray line of road, because the hand of man is set to the keeping of it open; but it crowds to the wheelruts, and in places where the pitch is steep and later builders have deviated from the straight line and made a curve so that the hill might be climbed more easily, it has swooped upon this untraveled bit and made forest of it again with amazing celerity.

That is the one astounding thing in this whole region of northern Maine,—the regenerative power of the forest. What could stand before the surgent growth of its young trees? Men with axes have been hacking at the giants of the wood up here for two centuries and more. The goliaths have been laid low indeed, yet for one tree that stood on a given space along the hillsides and in the valleys of Number One a century ago five stand to-day.

They are giants no more, it is true, but they are splendid trees; and just as the Liliputians might prevail where Gulliver was bound, so these trees hold their own against man and even press in on his clearings and wipe them out. There must be many more lumbermen with axes along the Macwahoc, the Moluncus, and the Mattawamkeag before this beautiful region will fail of its forest.

Over on the ridge, some miles to the westward of the Macwahoc-Kingman road, stands a sole survivor of the old-time pumpkin pines. Forty and fifty feet from the earth toward its limbs the birches and beeches lift whispering leaves. Timber and cat-spruce and resinous fir spire higher yet and fling incense toward him. Sixty and seventy feet they reach, growing tenuous to the tip of nothingness, yet the stately column of his trunk soars half a hundred feet beyond their tops, lonely and unapproachable.

It was to forests of such trees as these that our great-grandfathers brought their axes,—a forest that we unlucky moderns may see here in our dreams only. We are fortunate in having the stumps left, for they still stand along the Moluncus in much the same form that they stood when the lumberman's axe was yet pitchy with their chips. The roots are still sound wood, and it may be another half-century before they decay and add to the richness of the dense forest mold about them.

The stumps, five or six feet in diameter, and often as high as your head, showing in what depth of snow our ancestors worked at their logging, hold their shape in many instances. Around the base is a circular ring of dark rich mold which was once the bark on the stump. This has in every case fallen off and crumbled to humus, leaving

the heart-wood exposed. Mosses gray and green cling to this and cover it, and because it retains its shape you might almost think it sound, but a kick or a stab with your walking-stick will prove the opposite. It is but punk, standing in the breathless, windless silence of the wood, mute monument to a glory that is departed, waiting itself to pass on at a touch.

What the glory and solemnity of the Maine forest must have been when these giants were the columns to the temple of the woods we can but dream. In the dense shade of their dark, interlocking boughs no deciduous growth could thrive, and their own lower branches died for lack of sunlight and passed in time, leaving behind no scars to mar the splendid columns that rose fifty or sixty feet clear without knob or limb.

Out of these lofty, silent spaces must have stepped the tall gods of the red men, nor can one imagine the Indians themselves traversing them in other than silent reverence. Nor yet can we of a stronger race stand among their moss-grown stumps to-day without feeling the worshipful awe of the forest strong upon us. The gods are gone indeed, but the demigods remain. The spruces and firs, foster children of the great pines, stand close-set upon the ground that they once occupied and rear again the temple toward heaven in pinnacles and spires where once were darkly-vaulted domes.

You may worship here still, as I feel that you might have worshiped under the great pines, and I can but feel, too, that among the firs the wood gods are nearer and more gently kind than they may have been among the elder trees. The giant on the ridge, looming so high in cold reserve, seems too lonely and far away for human companionship. The spruces and firs are your friends, while yet the deep wood which they make loses no whit of its solemn nobility.

The timber-spruce, as it is commonly called, seems to drop its lower limbs a little more readily than its darker boughed brother, which goes by the name of cat-spruce among the local lumbermen, to thus prepare itself for the lumberman's axe as yielding a timber in which at a given age are fewer knots. White and black spruce, the botanists call them, they and the lumbermen definitely distinguishing between the two by minute differences, which to the new-comer in the big wood are not so easily appreciable.

You may know the fir more readily. It seems to me a tree of a finer, sweeter soul than any other evergreen. George Kimball, the novelist, who wrote "Piney Home" about the people who dwell among the quaint farms and silent stretches of interminable forests along the Moluncus and Macwahoc, puts it pithily and prettily when he says: "The spruces wear their hair pompadour; the firs part theirs in the middle."

The fir, indeed, is a Quaker lady among evergreen trees, with her hair so smoothly parted, her dark, unassuming, yet beautiful garb, and that soothing, alluring, healing fragrance which floats ever about her like an atmosphere of sincerity and loveliness. It seems as if all the wounds of all the other denizens of the wood might be brought to

her to heal, so loving is her presence, so benign the soothing influence that floats from her amber tears.

The sap of all trees has something of goodness and delight in it. The maples bear sugar that is more than sweetness; it has in it some Attic salt that makes the imagination smack its lips. The brew of the birch is more than beer; it is the embodiment of a flavor that bears dreams of rosy mornings on woody ridges that look down on the golden glory of the primeval world. So the faint fragrance of the fir floats like a divine presence from a loving heart that would fain clasp to itself the wounded and stricken of the world and dress their wounds and make them whole again. No wonder custom has adopted the fir for the Christmas tree. There is no tree so fit to bear loving gifts to all the world.

The spruce partridge, as he is commonly called up here, the Canada grouse (*Dendragapus canadensis*) of the scientists, is a bird that I find very common and amazingly unafraid under the spruce and fir in these northern woods. He is a smaller, grayer, darker bird than the ruffed grouse which is the familiar bird of our home woods. Up here they call the latter "birch partridge," because he feeds on birch buds, while the spruce partridge feeds on the tips of the spruce. The birch partridge is more wary. As at home, he thunders up from the underbrush and shoots himself across space and into the shelter of the farther wood like an indignant cannonball.

The spruce partridge winds along the brakes and undergrowth just ahead of you, or in the more open space under the dense evergreens flutters up into the lower branches, and seems to think himself secure there. I have stood among a flock of these beautiful creatures while they called faintly and reassuringly to one another,—so near that I might see every minute detail of plumage. Then, before they flew, I stepped quietly up and touched the soft feathers of the one on the lowest branch.

Then, indeed, panic fear seemed to



strike the flock at one blow, and they whirled into the dense green of some tender, motherly firs, whose arms closed about them and hid them from all rude intrusion. These birds are smaller than the ruffed grouse, though they are plump and beautiful creatures, and, because they feed on the spruce tips, are said to have flesh too strongly spiced to be palatable. I am glad of that. After the friendly way in which they received me into their community, to shoot and eat them would be a good deal like going out and bagging the neighborhood children on their way to primary school.

You soon get to feel that way about the deer up here in the Macwahoc woods. All along the lumber roads you may see their tracks, their keen hoofs cutting pointed marks in the soft mold of the wayside. If you have come silently and the wind is right you may swing a curve and be in time to hear a buck stamp and blow before he sees you and flips his flag and bounds off into the brush. Or you may see a slender doe pirouette like a ballet-dancing wood nymph and float away, with a stiff-legged, dappled fawn prancing after.

The creatures of the wilderness, when startled, seem to have a singular scorn of earth. You hardly note that they spurn it from beneath them as they depart. The coyote and jack-rabbit of the western plains do not seem to run; they simply float over the sage-brush, to your following vision much as a hawk does, only far swifter. So I have seen a fox sail along, seemingly about three feet in the air all the time, over a Massachusetts pasture. It is amazingly like flight. A startled Macwahoc deer in the same way seems to unconsciously acquire the true principle of the aeroplane.

In among the hackmatacks and arbor vitæ in the lower land the golden-winged woodpeckers are gathering in numbers in preparation for their fall migration southward. You may hear the vigorous note of the approaching single bird as he stops for a moment on a spruce top. "Kee-yer, kee-yer, kee-yer," he shouts, with the accent on the yer. It has all the loud nasal twang of the stage Yankee, and the bird is as ludicrously awkward in his ways, sometimes.

If you step softly through the swamp you may find a group of them going through a grotesque dance, seemingly for their own amusement. They spread their tails stiffly, walk along limbs with mincingly awkward gait, and bob and bow to one another, saying, meanwhile, "Wee tew, wee tew, wee tew." It is an amusing performance, and is apt to be interrupted by your guffaw of laughter, at which, whirls of white, gold, and black, with a dash of red, they fly away to repeat the performance in some undiscovered retreat.

The flicker, which is another of the fifty-seven varieties of alias under which the golden-winged woodpecker sometimes travels, is, I believe, the most brainy of the woodpecker tribe. Having brains he has also humor, and from the time he takes his first flight from the high hole in some woodland stub till pigeon hawk or barred owl cuts short his flickering, he is making a joke of things.

Like the flickers, the crows of northern Maine migrate southward in winter. The deep, long-remaining snows cover their sources of food too deep, and they find the clam flats of the coast a sure refuge and a well-stocked larder. Just now they are waxing fat on grasshoppers, marching in long lines across the open fields, lines from which no careless hopper may escape, and croaking contentment as they go.

They will stay until the snows drive them, however, and even in winter an occasional scout makes a quick flight north just to see how the land lies. It is but a half-day's trip up and back. I wish I might, too, be able to reach the land of the mother firs as easily when I feel the need of them. However, the aeroplane is in the incubator, and, unless the Wrights go wrong, perhaps next year or the year after I shall.

BIRDS OF THE NOR'EASTER

OUR weather here in eastern Massachusetts comes from the southwest. Whirling storms, little or big, move up from the Gulf coast and pass on, headed for Spitzbergen by way of Newfoundland. Knowing the habits of these whirling winds, the watchers of the weather bureau are able to say, as a rule quite accurately, when the storm will reach us, from what direction the winds will blow, and what they will bring with them and after them,—rain, gale, or fair weather.

One exception to this rule of accuracy is when the storm center, instead of reasonably and politely following the usual route, skips suddenly out to sea by way of Hatteras and goes roaring up the easterly edge of the Gulf Stream. That is when the weather signs that you find on the southeast corner of the front page, evening edition, fail, for that is when we catch our unexpected northeaster.

“Back to the wind in the northern hemisphere,” says the rule, “and the storm center is on your left.” So, with the wind whirling its thousand-mile circuit about this mysterious center halfway across the Atlantic, we get it from the northeast, and it brings whiffs of mid-ocean spume to our nostrils that are weary of the summer’s heat, and clothes all the land with the gray mists out of which grew the Norse sagas.

On days when the northeaster sings along the Gloucester shore, tears white wraiths off the red rocks of Marblehead and Nahant, and spins them in beaten spume along the gray sands of Nantasket, we of the inland country tread our heat-browned pastures with lifted heads, watching mysterious vapors wrap the land in legend, breathing the same air as the stormy petrel, and knowing that in our hearts the strong pulse beats with the blood of vikings.

On such days I love to watch the pond shore and the reedy stretches of the meadow marsh, for to them come the first of the wild migrants of autumn, and in the northeaster you may exchange greetings with the winter yellow-legs, just down from the Arctic shore. To-day I heard them, high in the invisible realms of the upper mists, whirling down to me,—gray forms out of a gray sky that seemed to loose them as it later will loose snowflakes.

Their staccato whistle in its minor chromatics shrills forth four notes over and over again,—notes lonesome with the heartache of northern barrens, wild as the echoes of ice cliffs that never rang responsive to voices other than those of the eerie birds of Arctic seas; a high-pitched plaint that might well be the shrilling of a little lost wind crying for its mother. You may imitate this whistle well enough to deceive the birds and bring them swirling within range of your gun if you will, though you can never put into it the wild plaint that echoes of far-off, lonely spaces.

The yellow-legs do not come as often as they used, and it is some years since I have seen even a small flock of the beautiful little blue-winged teal that were once so

plentiful that the rustle of their wings was a familiar thing at daybreak on the marsh. I miss them both. It is worth a tramp to pond or marsh to hobnob even for a brief moment of interchange of friendly greetings with such travelers. The winter yellow-legs may summer in the extreme Arctic and winter in Patagonia. The teal's range is less, though he may breed in Alaska and winter in South America. Their loss, here in the east, is the price we pay for civilization of our present sort. I daresay it is worth it, but I believe there is a better sort that does not come so high in the loss of wilderness friends.

Along the pond shore, after the yellow-legs have dashed in upon us, whistled the wind full of loneliness and heartache, and dashed away again like ghosts of gray snow-flurries yet to be, it is a pleasure to watch the homely antics of the spotted sandpipers. Of these you may find a pair or two about the pond all summer long, no doubt having a nest in some grassy meadow nearby. By the time the equinoctial northeaster is due, this pair or two has become oftentimes a dozen, preparing for their flight to the shores of the Caribbean Sea, where they will spend the winter, yet loth to leave New England.

These birds are never much afraid of me. If I approach too near they sing out peevishly, "Peet-weet, peet-weet," and half-circle in a short level flight out over the water and back again to the shore. Indeed, I strongly suspect their attitude toward my intrusion is one of humorous scorn. They are apt to face me as I come quite near, and bow low with what seems the exaggeration of politeness, only they immediately turn about and bow just as politely the other way, which flips their white tail feathers in my direction with a gesture which is certainly one of ill-bred contempt.

Then they fly away, leaving me in doubt as to whether they mean it or not. Probably, however, there is nothing distinctly personal in it. The legs of the spotted sandpiper are hitched to the body with muscles that seem to act like springs, and he can't help teetering when he attempts to stand still, hence his popular names of teeter, teeter-tail, etc.

Along with the spotted sandpipers at this time of year I am apt to find the ring-necked or piping plover, these already on their autumnal migration, for they breed from Labrador northward. They differ little from the sandpiper in size, but you will readily know them by the white collar which encircles the neck, with a little black vest partly defined just below it. Modest, busy little chaps they are, running about on the sands, picking up insects and minute *crustaceæ*, continually twittering "Peep, peep," and caring little for your approach until, finally frightened, they rise as one bird and fly away in a compact flock.

I have never seen these birds swim, though their half-webbed feet would seem to indicate that they can. Though, for that matter, birds that have no webbing at all between the toes sometimes swim well when forced to it. The common barn-yard hen, thrown into the water, will sit erect and swim as a duck might until her feathers are wet through.

To the pond with the autumnal northeaster usually comes a pied-billed grebe or two. If you are sharp eyed and fortunate you may see one beating his way down the wind with rapid strokes of his ludicrously short wings. His flight is something like that of a duck, though I think he makes harder work of it, more wing strokes to the minute; but you will know him as he nears you, for no duck ever stretched his head so eagerly forward or carried his legs dangling so far astern.

The bird should be at ease on land, for he has a bill like a hen, and his toes are lobed merely, not connected with webbing. But he is not. On foot he is slow, clumsy, and ludicrously ungainly. Probably for this reason the grebe does not go near land when he can help it. Even his nest is built on the water, sometimes actually floating, a mass of rotten sedge and mud, and the chicks swim and dive like old birds as soon as hatched. But if the land gait of the grebe is ludicrous and his flight laborious, in the water he is the personification of grace, ease, and agility.

Well does he merit one of his familiar names,—that of water-witch. When the hunters go forth to the marsh I am sorry for my innocent friends, the blue-winged teal. I know how few fly now where once the air would seem full of them. When I hear the quacking of live decoys my heart misgives me for the fate of the black duck, for I know how their fellow-feeling and sociable instinct will bring them in to the blind where the gunners are hidden.

Neither decoys nor dead shots give me any qualms of uneasiness where the pied-billed grebe is concerned. The decoys may split their throats in calling to him when they see him swim by just out of gunshot. He will not even turn his head. It may be that he has a voice; I have never heard him use it. When it is in the open with fair play, grebe against gun, my sympathies are with the gunner, for I know how great cry and little wool will result.

I have seen a pied-billed grebe cornered in a narrow, shallow river by gunners on each bank. He dove at the flash of the first gun, and though it was point-blank range, he was under water before the shot could reach him. He was up again and under a dozen times, to be followed by a dozen shots, only wasted. No wonder the hunters call him “hell-diver.” I have seen it stated in nature books that this name is given him because of the extreme depth to which he is supposed to go. No doubt the grebe goes deep when he wishes to, but the gunners haven’t taken that into consideration. The name is one symptom of the profanity which his exceptional skill necessitates.

At the end of a dozen shots the grebe cornered in the river decided in his slow way that he was being hunted while above water, so he simply failed to come up. A grebe has been known to stay under five minutes when loosening water-weeds for its nest or when pursuing fish for its supper. This one was seen no more by the gunners, and after waiting half an hour or so they went away, firm in the belief that the last shot had really reached him, but that he had in his death throes become entangled in water-weeds and remained there. Comforting for the gunners, no doubt, and very satisfactory

to the grebe. Ten minutes after they had disappeared the bird reappeared and went on feeding as before.

He had simply been floating along, under water all but the tip of its bill, which protruded as far as the nostrils and gave him ample opportunity to breathe. All these are clever feats, of course, but are explicable. The grebe has to live, either on or in the water, and he has learned how to do it even with the hand of man against him. He has one other trick, however, the mechanism of which I have never been able to understand. Swimming along on the surface he will, if he cares to, suddenly sink as if made of lead, feet first. How does he do this? One moment he is as buoyant as a cork; the next he goes down like a flatiron. "Spirit duck" is another name of his. He deserves it.

Another bird that is always linked, in my mind, with the sea wind beating the long marsh grasses into panicked waves and the fine rain of the equinoctial hanging the sheltered culms with strung pearls, is the Carolina rail. Some of them breed hereabouts, but the greater number of them are on their way from Labrador, where they have brought up the season's young, to the banks of the Orinoco or the steaming swamps that border the Amazon.

How they ever make the flight back and forth each year is one of those mysteries of which the wilderness world is fascinatingly full. Hardly with threats and beating of bushes can you drive them out of the marsh grass. When one of them does take to the wing it is with reluctance and apology for his awkwardness oozing from every pore. If you will put some brown feathers, a pair of dangling legs, and two short, inadequate wings on a misshapen bottle and send it fluttering through the air over the grass tops for a rod or two, you will have a good imitation of a Carolina rail protesting at being kicked out of the *Poa serotina*.

Once is always enough for him. You may go to the exact spot where he dropped into the grass again and raise all the hullabaloo you wish. Only with a dog can you start him out again, and the third time he will not flush even for the dog. Yet with this equipment *Porzana carolina* leaves Labrador in the latter part of August and arrives in Venezuela during November! Perhaps he does part of the journey on foot, for he is certainly better equipped for walking than for flight.

The rail is the incarnation of timidity, and you may look long even when the marsh is full of them before you see one. The best way is to slip your canoe quietly up some narrow creek where the tall grass waves far above your head and lie silent in it where you may scan either bank. Trampling through the grass it seems thick almost to impenetrability, but with your head on a level with its roots rather than the tops, you will see that it is full of Gothic-arched aisles, sometimes widening into under-grass cathedrals with nave and transept, sometimes narrowing into invisibility, though there is always a secret door through which the initiated may pass.

Down the widest of these aisles comes the runway of the muskrat. Through the tallest of them may stalk the bittern with his long neck stretched straight out before

him, and his sharp bill pointing the way. These are the broad highways of the marsh, but the rail does not travel them much. Even their seclusion is too public for him. He prefers the narrowing passages that lead him to close-pressed grass culms. These cannot bar his way, for that peculiar wedge-shaped build which makes him so ridiculous on the wing is just what he needs here. It allows him to follow the point of his bill and slip through the thickest growth of culms without a rustle and without disturbing the tops. Hence if you are fortunate enough to see him, he is just as likely to step forth from a solid wall of grass as from one of the pointed arches of the openings along the way.

You will not hear the grass rustle nor see it move, but the rail will be there, intent and preternaturally solemn. His head is thrust downward and forward, his tail is cocked nervously high behind, and he walks gingerly, as if apologizing to the mud for making tracks in it. You may see him climb a rush by clutching it with his toes, and feed on the seeds above; you may see him swim deftly across the creek, for he is a good swimmer. But the least motion on your part will send him into the thick grass again so quickly that he seems to dematerialize.

Old gunners tell me that a rail will slip under water and cling to a reed with only his bill above the surface, thus imitating the grebe in his methods of concealment. They say that when hard pushed by dogs and guns they go entirely beneath the surface and sometimes cling there until drowned; also that they have known rails to go into fits and finally swoon from fright. I cannot vouch for these things myself, but I believe that if any bird ever swooned from fright it was a Carolina rail.

Duck, grebe, plover, and rail may come to us storm-driven by the stress of the equinoctial. Not so the loon. He rides the northeaster, and you may hear him whooping in wild glee as he slides down the gale. His gray breast is brave to buffet gray crests of Arctic seas and his mighty thighs are built to drive the broad webbing of his agile feet till he whirls through icy waters like a spirit. Alert, defiant, mighty, he is a familiar figure of the wild gale that has spun a thousand miles across turbulent seas, and when he lights in our inland waters he comes not for refuge, but because the restless joy of storm riding has happened to bring him hither.

Shoot at him if you will. He is under, unharmed at the flash of your gun, and he may swim a half-mile, if he cares to, before coming up again. Then you may hear him laugh in scornful good humor, "Hoo, hoo, hoo, hoo," for little he cares for you. He knows enough to keep out of your way, but you cannot feel that he is afraid of you. When he goes out again, welting the gale with his strong wings and boring straight into the wild heart of the northeaster, the pond is lonely, the marsh flat and insipid, and it is time for dry clothes and the comfort of glowing logs in the wide fireplace.

THE SQUIRREL HARVEST

THE red squirrel is a good deal like me,—he never can wait for the chestnuts to open. As long ago as early September I used to see him going up and down the trunks of trees neighboring the chestnuts, sputtering and exploding his way along in a jerky unrhythm. He would go up the trunk as a light-weight, motor-skipping runabout goes up a steep hill, trembling all over as he fizzed along with barking explosions.

He had his eye on the closed burs, densely set with green spines, and he was angry because he was liable to get his tongue pierced in getting them open. But it did not matter. The milk-white pulp in the brown shells was too tempting. All this last month he has been going to the very tips of the limbs of the highest trees, clinging there as only a red squirrel can, and gnawing the burs loose. When a sufficient number of these were strewn on the ground beneath he would motor down there, and with the piston still chugging occasionally, just to prove to himself that he could start his car at a second's notice, cut expeditiously through the defiant prickles and smack his wounded lips over the kernels within.

Meanwhile, in common with most of the boys in town, I, too, have been having my troubles with the chestnut burs. A boy understands that the red squirrel gets the burs after the fashion of the real sport, and so far as he can he is willing to do the same. But the smaller limbs of the chestnut are brittle, and under the best of circumstances it is a



dangerous thing to go far enough out on them to reach the tips. Light-weight, daring boys sometimes do this, and often fall in the attempt, as accident records show. Sometimes the squirrel falls too, though this is of comparatively rare occurrence.

The wild creatures of the wood are as liable to accident as you and I, but they are not so prone to it. That severe pruning which wild life gives all who are robust enough to live it lops off all the clumsy branches of the squirrel family tree. Few but the cool-headed and skillful live to reproduce many of their kind.

The boy who falls from the upper limbs of the chestnut may save his neck by catching a lower limb as he falls—I have known boys to do it. Or he may even land with no serious injury if he is fortunate enough and the distance is not too great. The squirrel would be almost sure to land safely either in the lower limb or on the ground. This is more sure in the case of the red squirrel than in that of the gray, for the gray is two or three times the weight of the red. Yet I have seen a gray squirrel come down forty feet though the air and land uninjured.

My own method of loosing the unripe burs from their tenacious hold on the limb tips lacks the finesse of that of the squirrel. I do my work with a club. Nevertheless, it takes wisdom and precision. To stand twenty feet or so below a bunch of chestnut burs and hurl your club at them with such accuracy that it hits the limb just behind them at the right spot to snap them off their perch is an art that you must learn in boyhood or never.

You may hit the burs themselves or you may hit the limb farther back, and nothing happens. With the burs on the ground your task is to open them, which you must do by pounding with one stone upon another. Hit in the right place and with the right force, the green, prickly envelope yields and the soft, brown nuts roll out uncrushed. To me they are sweetest when this brown is just beginning to tinge them, before the shells are very hard and the kernel is too resilient and crunchy.

On these October mornings the chestnuts are ripe,—a wonderful rich brown, still clinging in close companionship in the center of the burs, which have opened and revealed the precious kernels within. To harvest them now by the quart your task is more easy than it was to get a few when they were three weeks younger. The squirrels know this. There is no need to climb to the dangerous limb tips and cling there precariously while gnawing them through. The ground is strewn with bounty, and the reds and the grays both are busy among the rustling brown leaves garnering what the winds, the boys, and I have shaken from the open burs and failed to gather.

Now and then they eat one, but for the most part they are busy storing them up for future use. In hollow trees, under stumps, they pile them in little hoards. But beside that they dig little holes in the ground here and there and put a nut at the bottom of them and pat the brown leaves down on top. I have always inferred that these were for special luncheons, stored ready to hand when the owner did not care to go to the main

larder. I know that they do go to these in the winter on occasions, for I have often seen the hole through the crusted snow where the squirrel resolutely dug his way in and left behind him the chipped shells of the nut which he found there. But I do not believe that one nut out of a hundred that is thus buried is ever resurrected by the squirrels; it is nature's method of getting her chestnut trees properly planted, and I half believe that the squirrels realize this; that they do not mean to dig these nuts up again, and only do so when hard pressed by hunger.

My path to the chestnut wood to-day lay through a shallow sea of purple wood-grass. It is a wild grass, scorned of the farmer and left ungarnered of his scythe, standing now in clumps in all waste places of the pasture,—an amber wine of autumn tint that intoxicates you as you pass through. It is a stirrup cup for your expedition. Old as the hills, amber-purple and clear, yet with a fine bubbling of hoary leaf tips, it warms the heart as wine of the grape does, and already you begin to be drunk with the beauty of the day. Afterward you pass through aisles of birch wood, where the once green leaves are a translucent yellow, fining the gold of the sunlight down to a soft radiance, a richness of pale effulgence that I have seen matched only in one gem.

Some years ago there came from South African mines a wonderful lump of crystallized carbon,—a great diamond that, cut and polished, yet weighed one hundred and twenty-five carats,—the famous Tiffany yellow diamond, in whose heart glows the same yellow radiance which wells throughout the birch wood of a sunlit October day. The Tiffany gem is worth its hundreds of thousands, and you might lose it from a hole in your vest pocket. The birch wood is a half-mile wide, and once you have felt its soft radiance flood your soul it is yours forever. Neither deserts nor cities can take it from you.

Sitting secure in a crotch of the chestnut tree of my choice, beating the chestnuts from the half-open burs with a birch pole and listening to their patter on the dry leaves far beneath, I was conscious after a time of a little gritting squeak,—a squeak that sounded much like a small, unoiled joint that was very mad about it. It might have been two tree limbs rubbing together, only that it was too personal. Creaking limbs are always mournful in tone; this squeak was full of impotent, nervous rage.

It was difficult to locate exactly, and I had thinned out the chestnuts pretty well and was about to climb down before I discovered what it was that made it. Hanging head down from a twig that protruded from the under side of a large limb was a great bat, swinging from one hind toe. His furry, gray body was half loosely wrapped in his wings, that looked like wrinkled folds of dark sheet rubber. His ugly little face was all screwed up with rage and his sputtering squeaks were a ludicrous exposition of impotent fury.

Every blow of my pole on the tree had jarred him. In his darkness of our daytime he could not see what it was that troubled him, nor could he venture to fly away from it lest he rush into worse danger. So he simply hung on and protested in all the voice and vocabulary that he had, and when I plucked him carefully by that hind claw and

wrapped him in a handkerchief and stowed him in the side pocket of my coat, he continued to mutter bat profanity.

You will find in the velvety heart of a chestnut bur usually three nuts, sometimes but one of these plump, and with a ripened kernel within the shell. The two others in this case will be but flat walls of shell with no kernel. Sometimes two of the three are meaty, and occasionally all three, only the fat ones being fertile seeds. Poking about among the brown leaves on the ground beneath the tree for these, now and then pricking my fingers in separating a particularly fat one from the bur, that had come down with it, I found another unfamiliar denizen of the chestnut tree that my clubbing had dislodged.

This was the larva of *Telia polyphemus*, the *polyphemus* moth. The moth himself is a beautiful creature with a six-inch spread of pinky-brown wings with a wonderful eye-spot of peacock-blue, dark-maroon, and yellow-white in the after wing. The form that I had picked up was a fat worm, nearly four inches long and fully an inch in diameter, of a clear, transparent, yellowish-green texture ornamented on the sides by raised lines of a silvery white,—a strikingly beautiful object so far as coloring is concerned.

The larva of the *Telia polyphemus* is no uncommon creature among oak and chestnut trees, although, so near is he in coloring to the leaves on which he feeds and so high in air does he spend his life, you may live in the woods for years without seeing one. Him I carefully stowed in another handkerchief, tucked into another side pocket, and started for home with my chestnuts and my menagerie. One more adventure, however, was in store for me.

In the open pasture stands a tall hickory, clad in the golden tan of autumn foliage, dripping gray nuts and blackened husks upon the pasture grass beneath it. Taking his pick among these was a splendid great gray squirrel, and as I approached, instead of bounding across the open to the thick wood, where he would have been surely safe, he sprang to the trunk, and hiding behind it, eyed me over the lowest limb.

There was something of roguish defiance in his look and I accepted the challenge. I dropped my coat on the grass, that the bat and caterpillar might be uncrushed in the mêlée and swung into the tree toward the squirrel, who promptly scampered up the trunk fifteen feet or so, poked his head over another limb, and undeniably winked at me.

The gray squirrel is clever, but even on his own tree his reasoning did not go very far. I was steadily driving him to the top, where he would be cornered, but he did not run out on a limb and drop to a lower one and then scramble down the tree and away, as he so easily might. He went straight on toward the top, and I after him. Hickory is tough, and even its small limbs will hold much weight. I could go as high as the squirrel could.

On the topmost bough he poised. I was within arm's reach. A gray squirrel has long, keen teeth and knows well how to use them in self-defence, yet you may grasp

one safely if you will do it right. Take him with the full hand from behind with the thumb and finger round his neck and meeting below his jaw. Thus you may hold him securely, uninjured, and be free from harm yourself. I have often pulled grown squirrels from the nest in this way.

But before my hand reached him the squirrel launched himself into the air with a bound that carried him in his flight clear of all limbs. It was forty feet to the drought-hardened pasture turf, and immediately I keenly regretted my frolic. A fall from that height, I thought, could but end in the death or injury of my friend. I looked to see him go to his finish, but he did nothing of the kind. Instead, he spread his legs wide, stiffened his tail, and fairly seemed to flatten himself as he went down, scaling to the ground instead of falling inertly, and though he struck with a considerable thud, he was up and scampering for the wood immediately.

The squirrel had won, though I can but think it was a foolhardy trick, and he would have done much better to slip down from tip to tip of the hickory limbs and circumvent me by circumnavigating me.

The crimson of the sunset lighted the path home with lambent radiance that made a twilight of the yellow glow beneath the birches and dulled the fire of the sumacs on the upland to a red as of dying embers. The purple wood-grass caught and held the complementaries of these fires reflected in its shadows till I seemed to stride through ashes of roses to the dun shadows of the lilacs in my own dooryard.

Here I bethought me of the bat, too long enshrouded in my pocket for his comfort, perhaps, and I unknotted the handkerchief, planning to slip him into an empty squirrel cage for a day's observation before I set him free. But I had forgotten that the sun was now below the horizon and that the bat could see as well as I could. Seemingly, he could see quicker, for before I could put fingers on him he slipped from the fold of the handkerchief, dove into the air, and with swift, sculling wings mounted over the tree tops and was away like the wind.

However, I had my chestnuts left, and my *Telia polyphemus* larva. Him I put in the butterfly cage without delay, along with some chestnut leaves, on which he might feed. He proceeded instead to spin himself a cocoon, rolling himself in one of the leaves in the corner of the box. There he will sleep lightly till spring, when I hope to see him come out a full-grown moth. I shall watch for him with much interest, for this species is very variable, and many aberrant forms and local races occur. There are even albinos, and melanic specimens also have been noted with the wings almost black.

AMONG AUTUMN LEAVES

THE deep woods catch all the rich colors of the autumn sunsets in their foliage. The dull reds and the vivid ones, the maroons and the scarlets, the golden yellows and the wondrously soft and mellow shades of tan and brown they hold till from a hilltop you see the forest afire. Flames flutter, embers glow and fall, and brown ashes and cinders remain.

Yet, if you walk far below the fire, in the forest aisles that are beginning to crisp under foot with the fallen embers of this conflagration, you are conscious of but one color sensation. A subtle glow pervades all things,—an atmosphere that is a yellow from which the sap has run, a very ghost of color. The domes of the hickories that grow in the open pasture are a rich brown, a most lovable shade; those hickory saplings that are rooted in the shade, and wait so patiently for fate to carry off the big trees that they may take their places, take an autumnal tint of this ghost of yellow also, and all the leaves of the wood ferns are pale with it,—a paleness that becomes with the more delicate an almost transparent whiteness.

We may ingeniously say that the reason that these leaves are so anæmic is that they grew in the shade and had not in their veins the good green blood of those that flourished in the open and absorbed from the sun and wind of summer the burn and tan that were to show in autumn. Yet, how can we be sure of this when those leaves which grow side by side on the same tree vary so in their autumnal tints?

Here upon a maple I find leaves that are still green, while others just beside them are scarlet. From the hilltop those maples which show the fieriest flame are the ones that on close inspection show leaves where the green and red mingle either in the same leaf or contiguous leaves. Perhaps the green, complementary color of the red takes the part of shadow background and throws up the more vivid color in greater prominence.

The swamp maples are unique in their way of taking on autumnal tints, anyway. In common with all trees that stand with their feet in the water, they lose the rich green of full summer growth long before the frosts touch them, and long before similar trees standing on upland slopes have any idea that autumn is approaching. Occasionally a maple branch growing on some swamp tree, bowered in a little cove of woodland greenery, will flame up in early July, as if some *ignis fatuus*, wandering in by ghostly moonlight from a near-by ditch, had touched the bough with strange fire that crimsoned but did not consume.

There is nothing the matter with the tree; it is well nourished and of vigorous growth, yet it flares this early signal that winter with her train is sooner or later to whistle down the tracks of the great northern road. Such a maple is like an over-zealous flagman who stands on the crossing and waves his signal before the train has even started from the distant city. I do not recall seeing this trait exhibited by other trees.

Again, individual trees of many species will show ruddy tints in the swamp, sometimes in early September, before other trees of the same species, standing near by, have even a suspicion of it. Yet this rule holds good; the swamp trees color first and lose their leaves first, the maples first of all. Sometimes by October first precocious specimens are bald, their gray polls conspicuous spots among the surrounding greenery. With their vivid colors, their premature baldness, their usually smaller size, and a generally devil-may-care air which, perhaps, is only seeming because of these facts, the swamp maples always appear to me like swashbucklers, roistering young blades in whom riots the wine of life, whose red faces early in the morning of the autumn and whose premature baldness both hint of dissipation. Their roots are deep in the richest of mold dissolved in the water of copious springs. The most bounteous of banquets and the warmest of wine is continually at their lips. It is no wonder if their youth is tempted to excesses.

Most of the lady birches stand aloof on the upland slopes; I notice not far enough away to forbid the handsome young maples from climbing out of their mire of dissipation to nibble the dry husks of gravel-bank breakfast food and drink dew among them if they have the courage. But not all thus withdraw in whispering groups. Down into the swamp others have stepped and stand, erect and dainty, among the rubicund roisterers. Social workers these without doubt, missionaries of the Birch C. T. U., who thus give their lives nobly to teaching by example.

Among the same temptations they stand, their shimmering green skirts drawn slimly about them, their slight forms erect, the very visible essence of virtue. The fervor of autumn touches them only with a pale-yellow aureola, which marks at once their freedom from taint of temptation and their saintliness. There is not much to prove it in a bird's-eye view of the swamp this October, yet I can but feel that these pure lives radiate an influence among the sensuous swamp maples.

Here and there you will find one of these the rich green of whose summer leaves turns to yellow hue at this time of year, though it is a creature-comfort yellow compared with the soft ethereality of the birches. Such, I believe, are on the road to conversion. The spirituality of their neighbors has touched them and they are beginning to be conscious of the beauty of temperate living and strive toward it.

Perhaps some autumn we shall note the presence of a great revival and the October swamp will be all one pale, misty nimbus of spirituality, a soft yellow radiance of saints who have spurned riotous living and glow with ethereal fires of renunciation. Then will the Birch C. T. U. hold a praise service.

On higher ground another maple which from its autumn coloration as well as other characteristics is a very near relative of the swamp maples is the white maple, sometimes called the silver-leaved maple. This, too, turns a vivid red in early October, though it holds its leaves a little longer than the red maples of the swamp. On the other hand, the imported Norway maples, more shapely and stately trees in their full growth than our own, line our streets and parks with noble round heads that are still green

except for a slight frosting of bronzy yellow on top, giving the tree a richness of dignified maturity that is beautiful to look upon. There is nothing of the missionary about these; they simply stand serene, placid reminders of the value of noble example.

Like these trees in the formation of symmetrical, rounded heads are the chestnuts, which are still green when the other deciduous trees of the wood have been caught in the conflagration of autumn coloring. Now, the first week in October being past, they show a certain yellowness of foliage which is enhanced by the yellow-brown of the ripe burs which throng the tips of their upper branches.

Twice during the year does the rich green of the chestnut leafage bloom with a richer tinting,—first in June, when the long staminate blossoms seem to pour in cascades from their billowed tops, and again at this time of year, when the ripened nuts push open the green burs of September and the falling sap leaves them at first a yellow-green and later a golden tan-brown. Walking beneath the trees to-day you are likely to get a rap on the head from a solid seal-brown chestnut, or even find your neck full of prickers where the fretful porcupine of a descending bur has jabbed you.

Already the ash trees, whose foliage has passed with much rapidity through olive-green and olive-yellow to tan-brown, which still holds a little of the olive tint, stand bare and gray against the sky, like the red maples, sure prophets of winter. The ash is never profuse of leaves. It drops them first of all in the autumn and is among the latest to put them forth in the spring. Even in the height of summer you cannot say that its foliage is dense; and when the slender brown leaves lie upon the ground they do not make a thick carpet. They merely crisp under foot instead of rustling.

Under a Norway maple the ground will later be half-leg deep in dense curled leaves that rustle and swish under your stride. You plough through them and they leap up and dance away from your progress, a splashing, undulating brown tide. Under oaks, much later, you find a similar sea, though its flood does not rise so high and there is a crisper rustle that is yet a large-hearted and generous sound. Under willows there is a silky crispness that is quite different from either.

So, blindfolded and led from one part of the forest to another, you might tell every tree under which you passed by the sound of its dead leaves under foot. So, too, knowing your tree, you might tell with accuracy the time of the year, the definite week of autumn in which your pilgrimage was taking place. Under the oaks to-day, though but a few leaves are yet on the ground, you would feel the round acorns under foot, and you would know that these were not chestnuts because of the lack of burs; so, too, you might know that you were under the white oak instead of the black by the different shape of the acorn.

If your foot-sense were not sufficiently subtle to note this difference—though if you were much addicted to life in the open woodland it would be—you still might, blindfolded, know the white oak from the black by the sweetness of its acorns. I sometimes think they are more pleasing to the palate than the chestnuts, though they have a slight astringency. Yet their meat is sweeter and, aside from the slight

bitterness, has more of flavor, as you will see if you will test first one and then the other. I think you will agree with me that the chestnut flavor is pale and insipid in comparison.

The black-oak acorn is a different fruit. Like the tree it seems to have absorbed all the bitterness of the wood. The white oak always seems to me to glow with the generous hospitality of the sunshine, the black oak to be morose and vindictive, a tree of dull days and shadow. I have little excuse for this feeling, unless it is because of their fruits.

The two trees grow side by side in the woodland, the black, if anything, the more vigorous in growth, yet the scaly whiteness of the bark of the one always seems hospitable, the rugose blackness of that of the other unfriendly. So with the fruit; the rich flavor of the white oak acorns is inviting, the meracious bitterness of the others is repellent. Out of the fact of this palatableness on the part of the one and repulsiveness on the part of the other has grown a singular condition in the southern states, where the trees as here once grew in equal profusion, side by side in the forests.

There it is the custom, and has been since the days of first settlement, to turn swine loose in the forests, where in the autumn they fatten on "mast," which is an old English name still in use there, but little known in New England. It means forest nuts of any kind, but especially acorns. These southern, forest-feeding swine have so loved the white-oak mast that they have in a large measure kept the trees from reproducing by eating all the seeds. The black-oak mast, on the contrary, they have rejected, as any wise animal would, leaving the seeds to be scattered about in profusion and reproduce more black oaks. Hence a scarcity of white oaks in southern forests where they would be welcome.

The oaks are more tenacious of their leaves than any other deciduous tree, though they are fairly early in showing autumn tints. Long after the reds of other trees of the wood are buried in the brown drifts that cover the roots from the too fierce frosts of winter the rich deep crimsons and red-browns of the oak remain. Indeed, the leaves of some species hold on all winter, and let go their grip only reluctantly when pushed off by the swelling buds of next spring's growth.

Their rustle, as they cling to the twigs in December, makes the wood vocal as the winter winds sift the snow softly down among them. Oftentimes before you see the first fine, far-apart flakes of the coming storm you may hear them pat here and there on a resonant oak leaf, and their presence makes the winter outlook more perfectly and comfortingly bleak as the fine flakes whirl through them. Snow amongst perfectly bare twigs fails of its full effect. You need the shiver of its sifting among the dry, persistent leaves of the oaks to realize all the beauty of its bleakness.

Now, however, the rich wine reds, the vivid crimsons, and the deep maroons that deepen on the one leaf into bluish purples and on the other into violet-browns mingled, as they are yet with the vigorous chlorophyl-green of the untinted leaf, these all are beginning to make up the more permanent glory of the full tide of autumn

color. Come with me, if you will, at sunset to the scrubby hill where three years ago the woodchoppers swept through like locusts, devouring every green thing that lay in their path.

They left behind them only gray stumps, dead limbs, and devastation. Yet hardly were their backs turned before the surgent vitality of spring swept upward from the earth-sheltered roots and burgeoned from the gray stumps in adventitious shoots that flushed purple with the excess of young blood in them. Four feet they grew, these new shoots, that year, and as much more the next, and now another forest of young oaks, black, white, red, scarlet, and scrub romps where the elder forest stood in majesty. Its leaves are fewer in number, but of enormous size and full of the riot of young life, with all the vigor of the parent tree sent up from the great deep roots.

Now their tide of sap is flowing back and the deep bronze-green is turning to the richest crimson and lake. Through these the golden radiance of the sun is drowned in a sea of bacchanal glory that makes the eye drunk and bewildered with its wine of crimson fires. To look toward it directly is to face a furnace of vivid liquid flames that makes the whole world green with flying blots of complementary color as you look away. Looking north or south to relieve the eye, you find that the rich color is still caught cunningly in the curves and facets of the leaves that glow like fire-rubies set in mosaics of chrysoprase, almandite, garnet, and carnelian. Turn again so that your back is to the sun and your eye rests among soft depths of umber lighted by rich reds that do not dazzle and flanked by tans and beryl. It is a world of glow and warmth and color that will long outlast the scarlets and yellows of the other deciduous trees, and even in the dead of winter the sunset fires will glow and flare in remembrances of these colors in the still-clinging leaves.

THE DAY THAT SUMMER CAME BACK

THE summer came back to-day, trailing gossamer garments over the pasture and adding the romance of August to the glamour of the mid-October woods. Where luminous purples hung deep in the shadows of the distance it painted them with a soft gray-blue bloom like that upon the grape. The undulating hills were as soft with it as if they were waves of the sub-tropic reaches of the Gulf Stream, where a wonderful film of purple efflorescence shimmers as far as eye may see.

The tan of hickories and the tawny yellow of chestnuts seem to break through this haze as the floating gulf weed does off Turk's Island or among the Bahamas, and when birds lift from the tree tops and sail away, it is as if a school of flying fishes were darting across your steamer's prow. The softly-breathing southern air is welling up from this mid-ocean river of mysterious romance and floating films of dreams all along our too clear-cut hills.

To-morrow the wind will be in the northwest again, the morning sun will glint on fields that are hoar with frost, and in the afternoon the Blue Hills will be blue no more, but brown with the rustling tannin of dead scrub oak leaves seen too clearly,—gray with granite angles, and sharply cut against a sky from which all dreams have fled. We had thought the summer too long and too hot, we welcomed the crispness and vigor of autumn, but to-day we walked abroad with joy in the warmth that again thrills us as with a fine touch of youth come back, and as little crinkles of heat shimmer upward from the brown fields we push forward, eager to bathe in it all once more.

All the out-door world seems dreamy with the same delight. The blue jays flutter back and forth on softer wing, and their usual strident clangor is subdued to an almost caressing babble, in which you think you hear the tones of spring love-making. They know the feel of nesting weather, and though it is but for a day it soothes them to happy response. This morning a robin, sure that spring had come again, sat up on the elm tree outside my window and greeted it with full-throated song, just as he had in June, and all day long there has been twittering of birds in the pasture and the forest.

Only a few of our host of summer visitor song birds remain, and the great wave of southward migration has passed us, yet to-day the pasture was vocal with the twittering of late passing warblers, and some even sang, *sotto voce*, to a sand-dance accompaniment of rustling leaves. The myrtle warblers were busy among the blue-gray, waxy, aromatic berries of the bayberry, which is their favorite food. The crop is good this year, portions of the pasture being almost blue with the close-set berries, and I think the myrtle warblers will linger long with us. Indeed, they have been reported as staying all winter when the bayberry supply is ample and sheltered from the worst of the north winds.

If they do the robins will stay with them, for the crop of cedar berries is a good one also. Almost all the red cedars have some, and some are so thick-set with them that

their bronze-green, now yellowing a little with the lessening sap, is all lightened up with an alluring blue. I do not blame the robins for lingering long with the cedar berries. I like them myself. They are a little dry, but very pleasantly sweet; and after the sweetness is gone there lingers on the palate a spicy aromatic flavor which is most enticing.

Some of our Norfolk County swamps are so thickly set with swamp white cedars that it is almost impossible for a man to push his way through their young growth. That north wind that can cut its way to the heart of these must be keen indeed, and here, when the berries are plentiful, you may find not only robins, but now and then a bluebird, and more frequently partridge woodpeckers, all winter long.

We had a killing frost only a night or two ago, the thermometer in sheltered positions marking twenty-five to twenty-eight degrees. It withered the grape leaves and took all tender things of the gardens and fields. Such a temperature for a long autumn night one would think would be death to those frail creatures of summer,—the butterflies. Yet to-day I saw a monarch soaring on strong red wings about the top of a great pine tree, sixty feet in air, seemingly seeking food among the resinous tips.

Across the fields a sulphur flitted his dainty way like a yellow fleck of animated sunshine. A few grizzled goldenrod and frost-bitten asters still bloom feebly for him, but in the swamp, undismayed, the witch-hazel twists its soft, yellow petal-fingers and sends out dainty perfume for his delectation. Over at the clubhouse a hunter's butterfly and two well-preserved specimens of the painted lady sunned themselves in warm spots on the shingles.

In spite of the summerlike quality of the day these seemed anxious. Now and then they fluttered eagerly about the building trying window fastenings and poking their heads into cracks, seemingly trying desperately to get in. They tried on the shady sides of the building as well as on the sunny, and though I cannot prove that it was not mere aimless wandering, it seemed to me to be done with a definite design. I think the painted ladies were hunting shelter in expectation that the day was a weather breeder. I think they knew that more cold weather was sure to follow, and though they had found shelter in which they were able to weather the first cold snap, they feared lest the next be too much for them, and hoped to get inside in some crevice next to a stove funnel.

Some butterflies, notably the *Antiopa vanessa*, which appears sometimes on warm days in February, winter successfully. Probably the *vanessa* is particularly resistant to cold. Probably also he has a peculiar faculty for finding shelter and safety, and I think the two hardy examples of *Pyrameis cardui* showed signs of some of the same instinct.

Later, in the full heat of the afternoon, when the thermometer stood at eighty degrees, I stood by the side of a long, straight country road leading north and south. One monarch butterfly after another was soaring along this road, seemingly not in haste, but making, nevertheless, a speed of six or seven miles an hour. And every one of them was heading due south on the trail of the one ahead, as if in a game of follow-

your-leader. Was the leader a wise old butterfly who had made the long southern road before, and were these others monarchs of this year's growth following him that they might reach the goal in safety?

Someone wiser than I may answer this, but if he does I shall ask him how he knows.

The *Anosia plexippus*, which is another name for the monarch, has fluttered about this road all summer long, never going outside his usual round from one flower clump to another. The cold snap of three days before may have wakened primal instincts in him and sent him on his southern migration, just as these may have set the *Pyrameis* to fluttering about the clubhouse, where there might be sheltered spots in which to try to pass the winter in safety. Or the compelling force may have been something entirely different. Who can ever know?

All along the borders of the swamp the witch-hazel is working out its peculiar and mysterious destiny. It is not this belated summer day, however, that has brought out its fragrant yellow blossoms. They unfolded just as cheerfully in the killing frost of three nights ago. Witch-hazel nuts are ripe now, the witch-faced husks splitting open and showing the glossy black kernels within, about as big as an apple seed, shaped like the enticing black eyes of the witch herself.

All among these nuts grow the scrawny blooms, sending out a delicate fragrance which is as soft and fragile as that of early spring flowers,—a refined and pleasing scent that brings a thought of far-away apple blossoms. Yet on this sunny day you may not catch this odor unless you put your face close to the flowers, for the vigor of the sun draws up the smell of tannin from all the dry leaves underfoot till the whole world seems a tea factory. Should the rustle of these leaves in the light autumn breeze be the silken swish of trailing Oriental garments, and slant-eyed people appear under pyramid hats and begin to gather them and pack them in chests marked with strange pencilings like those on the end of a red-winged blackbird's egg, I for one would not be surprised.

The blackbird himself is an Oriental mystic in disguise, and he marks the names of his children in Chinese characters round the big end of each egg. The next time you look into a blackbird's nest you notice if this is not so.

If you wish the odor of the witch-hazel blooms you must go to the swamp a morning after a showery night. Then the odor of the dead leaves will have been all washed out of the air, and the faint, fine fragrance of the latest flowers of the season flits daintily out to greet you as you fare down the path.

Yet, though flowers are rare on the third week in October and the pungency of dead leaves pervades the swamp, the upland pastures have a fine fragrance of their own,—a perfume so dainty and alluring that you look for its source in bewilderment, knowing that at this time of year no flowering shrub, no slender-blossoming vine, remains to float it down the wind.

It is not the pitchy aroma of the white pines. These have just carpeted all the floors of their house anew with last year's leaves. The new ones are not pitchy, and that resinous smell which the midsummer sun distills is hardly to be noticed in the wood. Nor are the pasture cedars to be thanked. Their prim, close-wrapping branches give forth a woodsy smell when bruised. It is not a perfume, and it comes only with turmoil. The soft southern wind bears no particle of it to your wistful senses. The hemlocks stand, beautiful but darkly morose, on the north side of the hill, and give forth no scent.

I searched the pasture long before I found it. Coming out from under the white pines into an open glade on the more barren soil, where the pitch pines begin to climb the slope, it always seemed stronger than anywhere else. It was as if rose-crowned Cytherea and all her attendant nymphs had just passed from perfumed baths and gone upward through the wood. If the soft moss had shown the heel marks of dainty sandals I should not have looked further. It was as possible that the garments of passing nymphs should have shed sweet odors on the glade as that these should float serenely there when all the flowers were dead. I paused among the pitch pines to consider the matter, and one of them thrust its branch tip directly into my face.

Then I thought I knew. The same fragrance emanated from the pitch-pine branch, stronger, indeed, somewhat more resinous, I thought, but practically the same. Six clubs crown the tip of every pitch-pine branch, one standing erect like a plume in the center, five arranged about its base at equal distances, not unlike a five-pointed star. These are the new shoots for next year, in rudimentary form to be sure, but all modeled carefully on what is to be.

There is the vigorous stem and the leaves as green as they will ever be again, indeed I think greener. The whole thing, which will be a perfect shoot a foot long, is compacted into a solid club less than an inch in length. Enclosing this is a fibrous husk which wraps it from all cold. Howsoever bitter the weather the life warmth of the young shoots is most carefully protected by this wrapping. But there is more than this. An air-tight, waterproof coating of hardened pitch is outside of the whole, completing an exceedingly neat, tasteful, and effective seal.

The pitch-pine mother trees have completed their preserving and now sit back and radiate perfume in satisfaction and kindly good will toward the whole world, for this slightly resinous sweetness does not come at all from the pitch-covered buds on the branch tips as I first thought. It seems to emanate from the whole tree. Cut a branch and take it home with you. Strip leaves and buds from it if you will; then smell the wood. It is there. But more than from anywhere else it seems to come from the mature leaves,—those which have borne the burden and the heat of the summer, and now are losing their rich green in a ripening which befits maturity and work well done.

All the evergreens take on this slight tendency to a mellow yellow as the autumn waxes. It is due, no doubt, to the lessening of the sap in the leaves. All winter they will

hold it, and when the joy of spring sends his lifeblood bounding back again, it will fade and leave them vigorously green once more.

Crossing the glade again on my homeward way I plucked branches of juniper so thickly studded with blue berries that there seemed scarcely room for the scaly-pointed leaves, and in so doing I stumbled upon the real secret of the dainty odor left by the goddess and her train. For the matured shoots and leaves of the juniper give off a fragrance that is as much more dainty than that of the pitch pine as that is more dainty than the strongly resinous odor of the white pine when cut or bruised.

Cytherea must have smiled upon the humbler juniper as she passed, and the dwarfed and stunted shrub must have caught the warmth of her eyes full in the heart, for it sits snug as the days shorten and radiates a happiness that is perfume, and sends the thought of the goddess to all who pass that way. The stronger odor of the pitch pine carries it far on the soft south wind across the glade and down the path through the pasture, but this is only the vehicle. The dainty essence of perfume which stops you as if a soft hand fell upon your arm floats from the loving heart of the rough and lowly juniper.

The sun of this day on which summer came back set in a pale sky that flushed with a tint of rose leaves, burning long before it died to ashes,—the cool, gray ashes of autumn twilight. Against this the slender tracery of birch twigs stood outlined delicately. Some leaves still cling to the birches, and these were silhouetted against the pale-rose glow in a soft haze that made a shadowy presentment of springtime all along the western sky. The year in its second childhood thus slips happily away from us in dreams of its youth. Through the August midday of the pitch-pine grove we pass to the home path among the birches, and though October dusk slips its cool hand into ours, it is only to lead us toward a western horizon where springtime seems still to wait for us wistfully.

WHEN AUTUMN PASSES

LAST night the superstitious leaves, forced to part from the home branch and begin a journey on Friday, knocked on wood as they went by, hoping thus to make a change in their luck, for the omens were all bad. The gibbous moon was peering over the eastern wood and they saw it over their left shoulders. Hence in their fall they turned round three times, still for luck!

They suspected also that they were being sent off in batches of thirteen and shivered lonesomely all the way to earth, where they scrambled together in groups and held their breaths, listening. Now and then one of them saw a ghost, and rustled the fact to the others, who took up the dreadful story with little spitting sounds of terror till all rose like a flock of frightened birds and shuddered into scrambling heaps behind tree trunks and in fence angles. They made the night eerie with their outcry. As fresh platoons came down the wood-knocking had the effect of xylophone solos, the dead march in Saul played by goblins in the lonesome trees that tossed their bare arms to the sky in mute grief.

All the out-door people seemed sorrowing, and more than half a prey to superstitious forebodings, for the passing of the hunter's moon marks the passing of autumn. November, it is true, is rated as an autumn month in the almanac, but I have no doubt that The Old Farmer knew better. He had to divide the year into four equal segments, and he did it very well. If November must be classed with either autumn or winter it belongs rather with autumn. But it simply ought to be classed with neither.

November is a month by itself, just as March is, and neither has more than the most casual connection with the season that has gone before. The year might better be divided into two seasons,—the one of growth, the other of rest, with November and March sort of dead centers, as they say in mechanics, interstellar space as they say in astronomy—voids between the two.

These wood-knocking leaves are the last from the elms. The native maples and ash trees were bare long ago, and though some of the still birches hold their yellow nimbus, many others are bare already. Only the oaks stand up to be counted with their rich crowns of red transmitting the sunlight till those at the right angle between you and the sun flash like fire rubies.

Yet, when I say this it is true only of the native trees of the forest. None of the foreigners hereabout seem to ripen up in glory or, indeed, to understand what a winter is before them and duly prepare for it. The purple lilacs of my garden hedge show a green that may be a little grimmer than it was in midsummer, but there is no hint of a ripening color in them nor have they lost a leaf. Their pith is trained to continental winters still, and though they have faced a half-century of New England cold, they still have the habit of the Persian uplands, which are their birthplace.

The white lilacs haven't even that dark green, but are a gentle shade,—almost like that of early springtime, when the leaves are hardly as yet half grown. The apple and pear trees have lost some leaves and others are browned by the frosts we have had, but none of those remaining show autumn coloring as we know it. They are simply darkened and grizzled. The Norway maples are showing a bronzy-yellow now, but holding their leaves bravely still, as if in the memory that, though the winter night of their homeland is long and dark, its shores are bathed by the Gulf Stream and the cold is late in coming. I think none of the imported trees and shrubs of Europe show the gorgeous coloring of our native ones, though they may have been here long enough to have been trained to it by the climate, if that is the cause of it.

Englishmen know nothing of the glory of autumn foliage until they come to America and see it. Then they are duly impressed, though you cannot always make them acknowledge it. Search English literature if you will, through prose and verse, and you will find no reference to any gorgeous reds and yellows of autumn. They don't have them. Thomson in his "Seasons" speaks, referring to autumn, of

“ ... a crowded umbrage, dusk and dun,
Of every hue from wan, declining green to sooty dark.”

It is a pity Wordsworth could not have been born in Cumberland County, Maine, instead of Cumberland County, England, and have tramped the hills of, say, West Mansfield, instead of Westmoreland, that our rich autumn ripening might have fruited in his verse. I wonder that the English do not plant our maples and our red oaks in their parks. It would be an interesting experiment to watch for fifty years or a hundred and see whether the trees changed to the English habit and lost their gorgeous hues, and whether, if they retained them, some English poet did not rise to the occasion and make them immortal in splendid verse.

Perhaps it would all be a failure. Our American men and women, transplanted, so soon lose their native characteristics and ripen, over-ripen in fact, into English men and women that there lurks with us an underlying fear that the trees might suffer from the insidious blight also. Perhaps it has been tried with the trees; it would be interesting to know.

I think the leaves were afraid to go home to earth in the dark last night, because it is rarely the custom of leaves to part from the tree in the night time. On still nights you may camp beneath a maple whose leaves have long glowed red and seemingly been ready to fall, and not hear a single spirit-rapping of falling leaf against limb. The frost may be white upon them in the morning, but not until the rising sun touches them will they loose their hold and fall to the waiting earth. Then with the kindly light upon them you may hear, if you listen intently, the little chirp of contentment with which they let go and flutter quietly down to their winter's rest. On a still frosty morning when the sun has first touched the trees these faint clucks make an infinitesimal chorus that is as sprightly as the morning light.

The xylophone ghost-march of last night was a far different thing. It came with little puffs of south wind after a bright, still day,—puffs that died out as soon as they had done the work, and left the night white and still under the gibbous moon. On all the leaves that had not scurried into shelter a white frost fell that filled them with ice-needles until they were crisp, and then sprouted miniature ghost-ferns all along their stems and upper sides.

Thus they lay stark until the white of the night gloomed into the gray of a daybreak fog that seemed to scatter all life in a formless void. After leaves have once been thoroughly frozen they dance about in the breeze no more. The forming and melting of ice crystals breaks up their cells and leaves them sodden and no longer elastic. They sag and sink and the chemic forces of the earth soon begin to work on them and resolve them into salts and humus that will go the rounds and form and nourish new leaves for another year.

You may see the ghost of autumn go up, these last mornings of October, in this dense white fog that often lingers late into the day. Last night was breathless with frost, after the leaves had done their ghost dancing, until the wan moon had begun to cushion down in the velvety blackness of the west and the gray of false dawn had stopped the winking of low-hung eastern stars.

The world was blank with silence. Until now, no matter how dark the night or how still, you had but to listen outdoors to hear the pulse of nature beat rhythmically, to hear the blood surging and singing through all her arteries. In that last hour before dawn the pulse had ceased and the blood stood stagnant. Then some outside presence held the mirror of the universe down close to the lips of the earth to see if she breathed. At first it was unclouded.

Then little wraiths of white mist shuddered up from meadowy hollows and others danced in bog tangle as will-o'-the-wisps might have done two months ago. These quivered together in soft gray masses that shut out the meadows and swamps, absorbing them and numbing them into a white nothingness. It was neither a rising tide nor a growth, but a sort of absorption. From my hilltop, in spite of the gathering darkness that seemed to be crowded together by advancing day, I could see the world gradually slipping back through chaos into the white glimmering nothingness of the nebular hypothesis.

On such mornings, even after the white light of dawn has filtered through this gray darkness and made its opaqueness visible, the world stays chloroformed. The keen frost chill which has endured until the coming of the fog is merged in the dense damp cold of this which goes deep. The frost chill just touches the surface and does not penetrate. It numbs your fingers or tingles your ears maybe, but it gives the blood a fillip that makes it dance merrily, and you are warm though it is cold. The fog chill works in your marrow and you are cold inside first.

I think the birds know the night before when one of these marrow-numbing fogs that wrap all the ghosts of autumn in their folds are coming on, for they seem to seek

closer shelter than usual in the heart of the evergreens, and even when the cold, gray light of dawn filters through the opaqueness they still resolutely hold their heads under their wings. There is no song on a morning like this, no cheery chirping even. They all know that they will get bronchitis if they try it.

The red squirrels are a little hoarse already; they have been caught by a little one earlier in the season and they have no mind to add to it. So they stay snug. They have made their winter nests now, often in the close, crinkly limbs of a large birch, often in a good-sized cedar that stands well among other trees, that they may have easy access to the squirrel highway. Some of them are in hollow trees and others still have taken a crow's nest for their foundation and have built a dome over it.

Wherever it is placed the material and architecture is the same,—a soft, silky lining of the finest shreds of the loose-hanging outer bark of the red cedar, wound round and round with coarser fiber of the same material, the whole making a round ball as big as a derby hat, or bigger, the walls being several inches thick. Entrance to this is by a round hole, just big enough for the slender animal to squeeze in from a convenient limb. The elasticity of the cedar fiber practically closes this hole after the squirrel has passed, and the family may cuddle together there snug through the coldest snap.

On a bright frosty morning you may hear the shrill pæan of the red squirrel ringing through the wood as soon as he can see. Then he is out and alert. On mornings like this when the chill fog hangs dense I never hear him, and I am quite sure he sticks close to his family, cuddled up in comfort in the middle of that warm nest.

The morning light breaks through such a vast cold cloud with difficulty, indeed we may not truthfully say that the morning breaks. Rather, it oozes, coming so slowly that without a watch in the pocket you would not know the lateness of the hour. By-and-by, if you watch the east carefully, you will be surprised to see how high the pale image of a morning sun is riding.

On such a morning few leaves fall. The chill dampness seems to revive their waning energies and they apply them to clinging just where they are. Perhaps the chill reminds them dimly that they still are protectors of next year's leaf buds that nestle close under most leafstalks and may be injured if the leaf is torn away too soon. These are well wrapped in tiny fur overcoats or resinous wrappers, to be sure, but I think, as the leaves seem to, that if anything could penetrate these clever coverings it would be one of these morning fogs which mark the passing of October.

But, though to us who stand at the bottom of the fog that ghostly image of a morning sun looks pale and impotent, its work is really vigorous and aggressive. Looking down on it from a sufficiently high hill we may see it shredding the upper surface into breakfast food and eating its way so rapidly downward that the rolling billows of mist ebb before its rays like a Bay of Fundy tide.

Long before mid-forenoon it has finished its repast. From below the fog seems to gradually grow warmer and to be dissolved in its own moisture. The frost that crisped underfoot before the mists began to shiver together in the lowlands now glistens as

dew under the yellow sun. The day warms toward the noon and we note with satisfaction what a perfect one it is. But not till the little winds of afternoon begin to bustle in among the trees do the leaves again begin to fall. The moisture is again dried out of their petioles and the xylophone solo tattoos once more the elfin tune to which they march on.

But now they do not go shuddering and in superstitious terror. Instead, there is a lilt to the music and they dance their way down. Some jig it alone. Others waltz cosily; but by far the larger number like best the sociable square dance and foot it in groups to the merry-go-round of the Portland Fancy. It is in such mood that we like best to say good-by to them.

NOVEMBER WOODS

NOVEMBER is Nature's stock-taking month, when she suspends her labors, stands aloof from her work, and counts up the dozens, noting them all on her list before she carefully puts them into the winter storehouse. To the very last of October her factory is still running, though on part time. By the first of December she has put things away.

November is the month in which she counts up the gain or loss and is happy or disconsolate, according to the tally. Why else these wonderful clear days on which you may see without a spyglass clear to the other end of your universe? On some of these days Nature smiles in delight over her success, and we say, this is the real Indian summer. She is pleased with the perfection and profuseness of the product. On others you will see her eyes cloud with tears, and sometimes a perfect passion of northeast tempest blots the landscape and drowns the world in a flood of rain. In this case she has discovered that the workers in some special department have been lazy or hampered by some unfortunate condition and their output is a failure.

There are years when the nuts do not mature and the squirrels must migrate or starve. On others the drought so dries the upland grasses that those of next year may not sprout as usual from the roots but must be propagated by seed, which of itself is scarce also because of the dryness. Or excessive rains so flood the lowlands that a thousand swamp and meadow products rot and write the word failure large over a whole department.

For Nature's successes are by no means easily won. She lays such plans for a hickory tree that if all the blossoms which open in May were to produce fruit the trees' tough limbs would be torn from their sockets with the weight of it long before maturity. Some years, because of storm or frost, the tree's crop is a total failure, but the resourceful mother, the moment she notes the death of the embryos, sets the wood to making a more vigorous growth than would have been possible in a fruiting season. Then, though she may weep in November over the loss of nuts, she will be able to smile through her tears at the thought that next year the tree will have far more ripe twigs for the bearing of nuts. Or the tree may produce a thousand nuts and the squirrels be too busy to plant more than a dozen of them. What is true of the hickory tree is true of all other creatures of the vegetable and animal world. Death stalks close upon the heels of birth, and a million fragile lives pass out unnoticed to one that greets our eyes in maturity. No wonder some years November is a month of wailing and Nature lets the storms of December blot the tally sheet with the white forgiveness of the snow before the almanac will agree that the month is half over.

The boundaries of the real month are thus not half so firmly set as that which the calendar proclaims. October may on the one end and December on the other so overlap it, some years, that Nature has hardly time for her bookkeeping. This year I

think November came a day or two earlier than the calendar figures it, for the last days of the calendar month of October went out with a perfect paroxysm of weeping.

Nature, even before she fairly got her tablets out for the tally, had a terrible pet about something. I think her grief must be because of the carelessness of man during the summer's and autumn's unprecedented drought whereby he has killed with his fires so much of the woodland growth. For other than this it seems to me that the year's work has been very successful. Never were wild fruits more plentiful. Only on the driest of the upland pastures was there failure. There the fruit set in more than the usual quantity, but in some cases shrivelled before coming to maturity.

There was a tremendous crop of chestnuts this year, with enough hickory and hazel nuts to make the squirrels smile and work overtime in laying them up for the winter. From the June berries which purpled the shad bush to the wild apples that still hang on the woodland trees, gleaming pale-yellow among the rugged tracery of bare branches, production has been plentiful and picking peaceful. Hardly a rainy night, never a rough storm, did we have from the first of May until the end of September. All those trees whose fruiting depends upon windborne pollen which can only float in dry weather had perfect conditions for fertilization. So with those plants, whether shrub or tree or annual or perennial herbs, that depend on insects for the same service. There was no time lost on account of rain.

As it was in the vegetable world, so it has been with animal life, and particularly with those birds which nest on the ground. The mother bird may conceal her nest so carefully that neither skunk nor fox nor predatory boys can find it. She cannot conceal it from the rapidly-rising water of a June flood which will drown her nestlings or so chill her eggs that they will fail to hatch. A long heavy rain at just about hatching time may almost wipe out the young birds of a season among certain varieties. I read recently a report from Maine stating that the partridges are particularly plentiful in that State this year. This, the report went on to say, was because the hedgehog bounty of some years ago had made a scarcity of hedgehogs. Therefore, as the hedgehogs no longer ate the partridge eggs, partridges were increasing in number.

The State of Maine porcupine, commonly called hedgehog, though purists decry the custom, will eat the handle off your canoe paddle, the floor off your camp, or the boots off your feet. I dare say he eats partridge eggs when in his short-sighted, clumsy wanderings he happens to find them, but I doubt if he does enough of this to make him responsible for a shortage in the partridge crop. I believe the partridges are particularly plentiful Down East this year because there was never a cloud in the sky nor a drenching rain from the time the eggs were laid until the young birds were fully fledged. I know that is what happened here in Massachusetts and, as a consequence, the young of ground-nesting birds have had more than their usual opportunity to grow up.

This is true of partridges, and the application is apt, for the partridge is not a migrating bird, nor even a wanderer. He clings to the particular section of woodland

where he was brought up with a faithfulness which is apt to prevent his reaching a green old age. You may drive him from his covert with all the racket you are able to make. He may leave with vigor and directness that would seem to prove that he has through tickets for Seattle. Yet, if you sit quietly by in a position which commands a good view of the approaches, you will before long see the flip of a brown wing that is bearing him back again. He has gone no farther than the dense shelter of a neighboring pine grove, whence he watches out until he thinks it safe to come home.

I take it that the same reason holds good for the plentifulness of woodcock this fall in certain swamps which I frequent. You may know that woodcock are plentiful in a place, even if you do not see them, by the numbers of little round holes in moist, soft ground, usually where the swamp begins to give way to sandy upland. Here the bird goes jabbing for angleworms, which are his chief diet. I have never been able to catch them at it, though I have often noticed the borings in the spot whence I have just flushed the bird. In fact, I have never seen a live woodcock on the ground anyway.

The bird is so built that I and other predatory creatures will not be able to do it. His coloring is well adapted to blend with the dusky-browns and black of the low ground which he frequents. He does not have to look for his food. He feels for it. Given the proper piece of ground to contain angleworms, he has but to probe with that long, sensitive bill and haul them out when the sense of touch tells him that one is there. For



He does not have to look for his food

this purpose the end of the upper mandible is somewhat flexible and moves so as to nip the worm when he feels it.

If we could see him thus engaged I think we would understand clearly why a woodcock is so peculiarly built. His eyes are set so far back in his head that the bird has a grotesque appearance. But in this very fact lies a large factor of his safety. Wild animals that hunt woodcock may not slip up on them unseen while they are feeding. The woodcock's nose may be in the mud, but his eyes, set absurdly far back on his head, are then just right for seeing all that is going on. Let there be but the slightest hint of danger near by and the bird goes straight up in the air in a tremendous burst of speed.

Woodcock hunters claim that this speed is so great that the bird is invisible till he reaches a height of four or five feet. I am inclined to believe them for I have never yet seen a flushed bird till he got shoulder high, though he may have come up right in front of my nose. So vigorous are the strokes of his wings during this flight that the stiff wing feathers make a shrill whistling which is peculiar to the bird. Rapidity of flight seems to be in the main exhausted by this effort, however, for after they get fairly launched they seem to go rather slowly and clumsily. In the case of the woodcock, as in that of the partridge, the rainless spring and early summer seem to have given the birds a chance to bring their full complement of young through to maturity.

So, looking over the result of harvest and round-up in pasture and woodland, I can see no reason why Nature should shed many tears or go into any tantrums over the results of her busy season. These seem to me to be above the average, and I look forward to a bright and sunny November, during which she will count up the finished product with all good cheer.

The tally of young brought to successful maturity is all that the animal world has to show for the success of its department during the season of growth. But nuts and fruit and ripe seeds are only part of the work of the trees and shrubs. All the time that they are busy producing that two feet or less of woody growth, all the time the growing and ripening of seeds is going on, there is a further and very important labor to be attended to. That is the production of next year's buds. This is no haphazard matter, nor is it left until the other things are out of the way, but is carefully begun and patiently carried on through the summer, early autumn seeing everything complete.

The falling of leaves and ripe fruit shows these hopes for future foliage and flower revealed for the first time. Stand on a knoll in the pasture and look over the tops of shrubs and trees on these keen and clear November days and you will see that the most beautiful colors of the year are there waiting your eye after you thought that all color had flamed to its climax and died in the dead ashes of autumn memories. Grays that are incredibly soft and coot in the vigorous young limbs of the maples warm into tender reds on the twig tips where the next year's buds sit snug.

All this year's shoots of the swamp blueberry bushes are a restful green, but at the tips these, too, ripen into red, while on the higher ground the black huckleberries and the birches show the same color till the landscape rolls away from you in a warm and cuddly glow that takes the nip out of the wind. Looking on these you know that the pasture cannot be cold, however deep the snows to come or however low the mercury in the thermometer may fall. As the winter comes on this blanket of warm red, spread all over the bare trees and shrubs, will deepen in hue and with the first promise of spring flush into a lively pink that melts again into slender green with the passing of frost from the roots and the first soft rains of April. Herein is the better half of the harvest of the year,—a harvest not of fruition but of promise. The out-door world ripens hope in the same crop that has given us fulfilment.

How full of hope, of promises, of matured plans and energy these rosy buds are you may not know till you step down among them and test their virility and perfection. Here is the azalia, its pinky twigs tipped with swollen, soft green buds as big as your little finger tip. Till the leaves fell nobody thought the azalia had been doing anything since its rich-scented white flowers fell last July. Here is the proof of its labors and foresight. In the hearts of these buds are next July's blossoms, in miniature it is true, but perfect in every appointment.

About them are the green young leaves, vividly colored already, both only waiting for the mysterious thrill of spring sap to push forward to maturity, promising the leaves softly green, the blossoms vividly white, sticky with sweetness, and adorably fragrant. If you will pull one of the larger of the azalia buds apart you may easily see all this, and as you do it, be haunted by the ghost of a perfume, an infinitesimally faint promise of the rich odor yet to be.

So, in large or small, it is with all the shrubs and trees. Each is loaded and primed and waits but the touch of the match in the crescent warmth of the spring sun. Then will come the yearly explosion. It is hard to say which of these next-year promises shows most vigor, yet I think on the whole I would give the prize to the sapling pines. Each central shoot of these will go up in the season from fifteen to thirty inches, and send out four or five laterals. Yet each young tree has from eight to a dozen brown buds prepared for this, at least two centrals which you will recognize as being larger and standing more erect. One of these will get the start and continue the main trunk of the tree. The other will fall back and be a lateral branch. Yet if, as often happens, the central shoot is disabled the next strongest will take its place and so on, if need be, till the last of the dozen buds has stepped into the place of the lost leader.

Sometimes, though rarely with the white pine, more often with the fir and spruce, two will compete with equal success for this lost leadership and you have a tree with twin tops. Usually, however, one fails in the race and the stronger goes ahead alone.

So, going abroad these keen November days, looking upon the world stripped of the glamour of summer and the glory of autumn fruitage, we see it by no means a dead and pulseless thing to be wept over and buried. Instead, we wonder at and delight in

the riot of life laid bare by the passing of leaf and fruit. The woodland is more beautiful, the pasture more enticing than ever. Beauty thus unadorned is adorned the most, and we forget to sorrow over the ceasing of this year's growth in our joy in the promise of that for the year to be.

WINTER BIRDS'-NESTING

LAST night the world was all soft with mist. Over on the brow of Cemetery Hill you looked off into an illimitable distance of it. Horizon after horizon loomed over the shoulder of its fellows as the gray-draped hills rose one beyond the other and tiptoed softly away into the yonder world,—so softly that you could not tell where the earth ended and the heavens began.

The landscape passed like an elder saint from this world to the next, you could scarce tell when, only that you were awed and soothed with the soft serenity of the going. In the hush that followed the soft blue mists changed their draperies for black, in mourning for the passing of the twilight saint, and thus night came.

Last summer night on this hilltop was filled with voices. A million insects chirped and sang. Tree toads trilled, amorous toads played bagpipes all along the margin of the swamp below, and in deeper water a thousand frogs shouted one to another in guttural diapason. A little screech owl used to sit in the darker corner of the pines and ululate all to himself far into the night, and here and there a songbird, stirring in his sleep, would pipe a mellow note. A coon would whinny or a fox would yap, and there were many other sounds whose source you might not surely define. The forefathers who wait serenely beneath their slate headstones all along the brow of the hill had much and pleasant company when the year was in its prime. Now their nights are as silent as if the world itself were dead, their company ghosts of mist as tenuous as their own.

The morning after such a night does not break from above; it grows. It rises out of the earth like a soft tide, as if the mists that went to sleep in it last night were the first of all creatures up, making all things gray again. These tiptoe up, tangling their soft garments in the trees and roof tops till they slip from them and pass on into the upper spaces, where their unclothed spirits become the morning light. The garments, clinging still to all things, remain behind as hoar frost.

That is the way it was this morning. All the trees had white baby leaves of infinite daintiness and ghosts of blossoms that were not real enough for a promise. I might better call them remembrance, touched with hope. Hardly was the touch of hope there at the earliest light. It was just white and delicate remembrance. Then, with the thought of the sun, only the thought for the sun himself was not to come for long, there came a slender opalescence welling through these white garments, an iridescent presence that you felt rather than saw, till I knew without looking to the east that the dawn had grown out of the earth into the high heavens and the miracle was complete.

Out of this miracle of the birth of morning light came two pleasant things. One was the red sun, peeping robustly in among the pines, adding his glow to the warmth of their shelter; the other was a bustle of merry company heralded by a salvo of elfin trumpets. A company of chickadees came breakfasting, and with them were

nuthatches. I think no one has ever see the trumpet which the nuthatch blows, but its tiny, tin toot is a familiar sound in the pine woods at this time of year.

If some fay of the fairy orchestra, returning in haste from revels which lasted till the gray of the morning, did not drop it, I cannot tell where the nuthatch did pick it up. Its note is certainly more elfin than bird-like and always seems to add a tiny touch of romantic mystery to the day.

Such a November morning is fine for birds'-nesting. You may go hunting birds' nests in June if you wish to, but you will not find very many, half so many in a day as I can find now almost in a glance. Down stream there is a little island crowded with alder and elder, milkweed and joe-pye weed, and garlanded with virgin's bower, where I called many days last summer to watch the insect life that rioted about it. A bed of milkweed bloom was each day a busy and cosmopolitan community.

Right at my elbow as I stood in July watching this was a blackbird's nest. I must have brushed it more than once, but I never saw it until to-day. To be sure, when I first went there the young blackbirds were grown up and gone, for the nesting season with these birds is short, and by July the young are flying about with the flock, learning to sing "tchk, tchk, conkaree." Had there been young or eggs in the nest the distress of the parent birds would have warned me of its presence. Lacking that, so cleverly was it placed for safety and concealment, I never noticed it till the passing of the leaves left it bare.

Ten feet away was another, a replica of the first. Among blackbirds good form in house-building has but one accepted style. The nest is rather deep, loosely woven of rough grass, lined with finer grasses. Standing on the little island to-day I could not help seeing these two nests which before I had passed a score of times without seeing, for if June is the time of year to hunt for birds' nests, this is the time of year to find them.

The birds can give you, and I really think they are right about it, many reasons why you should not hunt for their nests in June. Looking at a nestful of young birds, with the mother fluttering solicitously about, I always feel as I think I should if I went into a town where I was not acquainted and went about peeping in at the nursery windows of peoples' houses. My motives might be the best in the world. I might be making a study of nestlings and nests of the human family for scientific purposes; in fact, I might be a veritable "friendly visitor," but I should be fortunate if I did not fall under suspicion, become the object of dislike, and eventually land in the police court.

The mere too frequent inspection of the nests and eggs of some birds will cause abandonment, and those parents who stand by do so with such evident distress that after the briefest possible satisfactory inspection we ought to apologize for the intrusion and step away. Many birds will even attempt to hasten this departure by pretty vigorous means.

None of these objections obtains now. There are no birds in this year's nests, and you may gather them or tear them to pieces in analytical mood without doing harm, at

least to the birds. Down stream, ten feet from my second blackbird's nest, was a catbird's. The catbird builds a better nest than the blackbird, at least so far as strength is concerned. Before the winter is over the grasses of the latter's structure will be broken and blown away by the wind or washed back to earth by rain and snow.

The catbird's will surely stand until next fall, and remnants of it may be sometimes seen in the bush the year after that. For the catbird's material is of more rugged quality. His foundation is often of pliant twigs or tough bark of the wild grapevine, though the nest I have before me as I write—the one which I could not see last summer when I passed it at the foot of the little island—has strong, coarse grasses loosely interwoven for its foundation. Then, within this loose, rough cup is a layer of tough oak leaves, the dry ones of the year before, wind-proofing the bottom of the structure. Then comes a layer of fine black roots, I think those of alder, taken where the stream had washed them bare. Then more oak leaves, and finally an inner lining of finer black roots from the same source as those already used.

The whole is firm, sanitary, wind-proof, but not air-proof, and sufficiently cup-shaped to hold the young securely, though not so deep as that of the blackbird. One kick would smash a blackbird's nest to a handful of straw. You might kick a catbird's all about the meadow, and I am quite sure the inner structure would remain interwoven.

I think the reason for the difference in the two is this. Though both often build over water and in similar situations, the blackbird has but one brood a season, and even a frail nest will do for this. The



A field mouse had appropriated this nest for an autumn storehouse

catbird hardly has his first brood off the nest before preparations are in hand for a second; and the nest which can stand two broods of riotous youngsters in succession, even if fixed up a bit, must needs be of fairly firm texture.

The strength of the catbird's nest often serves another purpose, though I doubt if this is taken into the calculations when it is planned and built. I found one of the half-dozen which line the brook conspicuously, now that they may be seen at all, half full of wild cherry stones. Evidently a field mouse had appropriated this nest for an autumn storehouse, perhaps planning, before the weather got too cold, to roof it over with a dome of soft grasses, this work of the field mouse being not so very different from that of the red squirrel, only on a smaller scale.

Farther down stream in a rough portion of the pasture, brambly but beautiful with barberries, is the chosen habitat of the yellow warblers of my neighborhood. Always they build in the barberry bushes here, nor have I ever found them anywhere else or in other bushes. It is not difficult to find them when the pasture is in the full leafage of late May, for you have but to go from one barberry bush to another till you have succeeded. But the yellow warbler is a shy bird, and I have known them to desert nest and eggs when these were too often visited.

It is much better to hunt them now, when you have but to stand on a little hillock and count, then pluck the nest that you prefer and take it home with you without abridging anybody's feelings. The yellow warbler mother bird seems to have a great love for the tender buff wool of the young shoots of the cinnamon fern, which are just about ready to shed these delicate overcoats when nesting begins with the yellow warblers. In fact, her color scheme is perfect.

The nest, when finished, is a symphony of pale buff and silvery grays that shade imperceptibly toward the buff touches on the under parts of the warbler and are lighted as with a gleam of sunlight by the bright yellow of the remaining plumage. Yet this bright yellow has a greenish tint that is deepened in the tender green of the young shoots of the barberry, while the yellow itself is again reproduced in the blossoms. No wonder this lovely little singing-bird loves a barberry bush for its nest. It finds protection and an artistically satisfying color scheme in the same bush.

The silvery grays of the nest are the fine, silky, fibrous inner bark of the milkweed, whose last year's stems are shredded by wind and storm in time for the nest-building. These barberry-bush-building yellow warblers with whom I have been more or less acquainted for a quarter of a century seem to care for little else for material, though sometimes they make the fern fuzz more adhesive with caterpillars' silk and line with a few horsehairs and soft feathers.

Yet though these nests have been invariable in material they have varied otherwise. Some have been so firmly woven and the material so stoutly packed as to defy the storms of a winter or two. Others have been so frail as hardly to be found when the

leaves are off. Perhaps these slight nests are made by birds that were nestlings of the previous year and have not yet learned the complete art of nest-building.

Once I found one whose makers were skilled indeed. Instead of placing it firmly in a crotch and building up with the fern wool within a netting of fiber wound from twig to twig, as is the usual method, these had launched boldly into a new architecture. Perhaps they had neighbored the year before with a vireo. Anyway, they took the vireo's plans and built a yellow warbler's nest on them, hanging it from a nearly horizontal barberry fork, and finishing a fine, firm, pensile nest, vireo style, out of yellow warbler material. I never found this nest's successor, and I am not sure whether, having found they could do it, they abandoned the type for the old home style, or whether something happened to the birds, and thus the warbler world lost budding genius.

Only one other nest have I found that seemed to be in any way abnormal, and this, unlike the pensile nest, seems to have had a very definite reason for its abnormality. The hollow part which had contained the eggs and young was in no wise different from that of all other warblers' nests. It was the depth and firmness of the foundation which surprised me. This was built up to the height of an ordinary yellow warbler's nest before the real nest began at all, and (the young had flown) I promptly took it home and dissected it.

Then the murder was out. The extra height had been added to the structure to circumvent the villainy of a cowbird. The cowbird lays her eggs in nests of birds that are smaller than herself and there leaves them to be hatched. She is partial to yellow warblers' nests because the eggs that belong there are much like hers in coloring, though smaller, and the fraud is less likely to be detected. When hatched the young cowbird is so much larger and stronger that it starves out the other nestlings or crowds them out. The nest-builders in the main are foolish enough to bring up this murderous changeling; hence cowbirds are perpetuated. Perhaps these warblers had had one experience.

Anyway, finding the cowbird's egg in their nest, they had promptly roofed it over with fern wool and fiber, built up the sides to correspond to the addition, and gone on with their housekeeping. Here was evidence of prompt action in an emergency in nest-building. I do not think it possible for the birds to have lifted the cowbird's egg over the side of their nest and to have dropped it on the ground, which would have been the quickest way of getting rid of it. A yellow warbler's nest "tumbles home" a bit at the top, as does the hull of a yacht, and I do not think their slender claws could grasp the egg and get it over that lip. Instead, they had done what they could,—imprisoned the intruder egg where it could not hatch.

I found it there, addled and nearly dried up within, and I rejoiced. The cowbird is a light-o'-love and abandons children on other people's doorsteps. All such should be put in a pie. Since English sparrows became so plentiful the cowbird has shown a decided partiality for their nests for its abandoned offspring. I found a cowbird's egg

with those of an English sparrow that nested in a crevice right over my front door last spring. If cowbirds must behave in this nefarious manner it is not so bad to find them choosing the English sparrows for their dupes. The surprising part of it is to find the cowbird with sufficient courage to come in under the porch.

I'd like to watch a young cowbird growing up in a nestful of young English sparrows. The tender nestlings of the yellow warbler have no show, but I have an idea that here Greek would meet Greek, and after the tug-of-war the cowbird would be among those not present. Perhaps in the falling out both would fall out, at which most of us who love birds would not grumble.

SOME CROWS I HAVE KNOWN

ALREADY the robins that piped such a deafening morning chorus all about us last June are swirling in great flocks about the Florida everglades, getting up a Christmas spirit by filling their crops with holly berries and practicing spring songs, and perhaps a little spring love-making in the waxy shadows of the mistletoe bough.

But not all of them. Yesterday, at sunset, I heard one that had not joined the innumerable throng. Instead, he lingers to take his Christmas dinner in New England, his holly the red-berried alder, his mistletoe black instead of white, with the crowded fruit of the buckthorn. Like his mates, a thousand miles away, he, too, sang a faint little winter song that was like an echo of his summer jubilate, a triumphant, light-hearted tune indeed, but not heartily sung. Twilight gloomed the deep pine growth where we were, and though the fires of a November sunset burned red and angry in the sky, they warmed the grove only to the eye, while the keen north wind that had blowzed the sky with clouds all day seemed to be seeking shelter there with us. He, too, whistled in such keen sibillation that the faint oak-leaf rustle from the hillside sounded like chattering teeth.

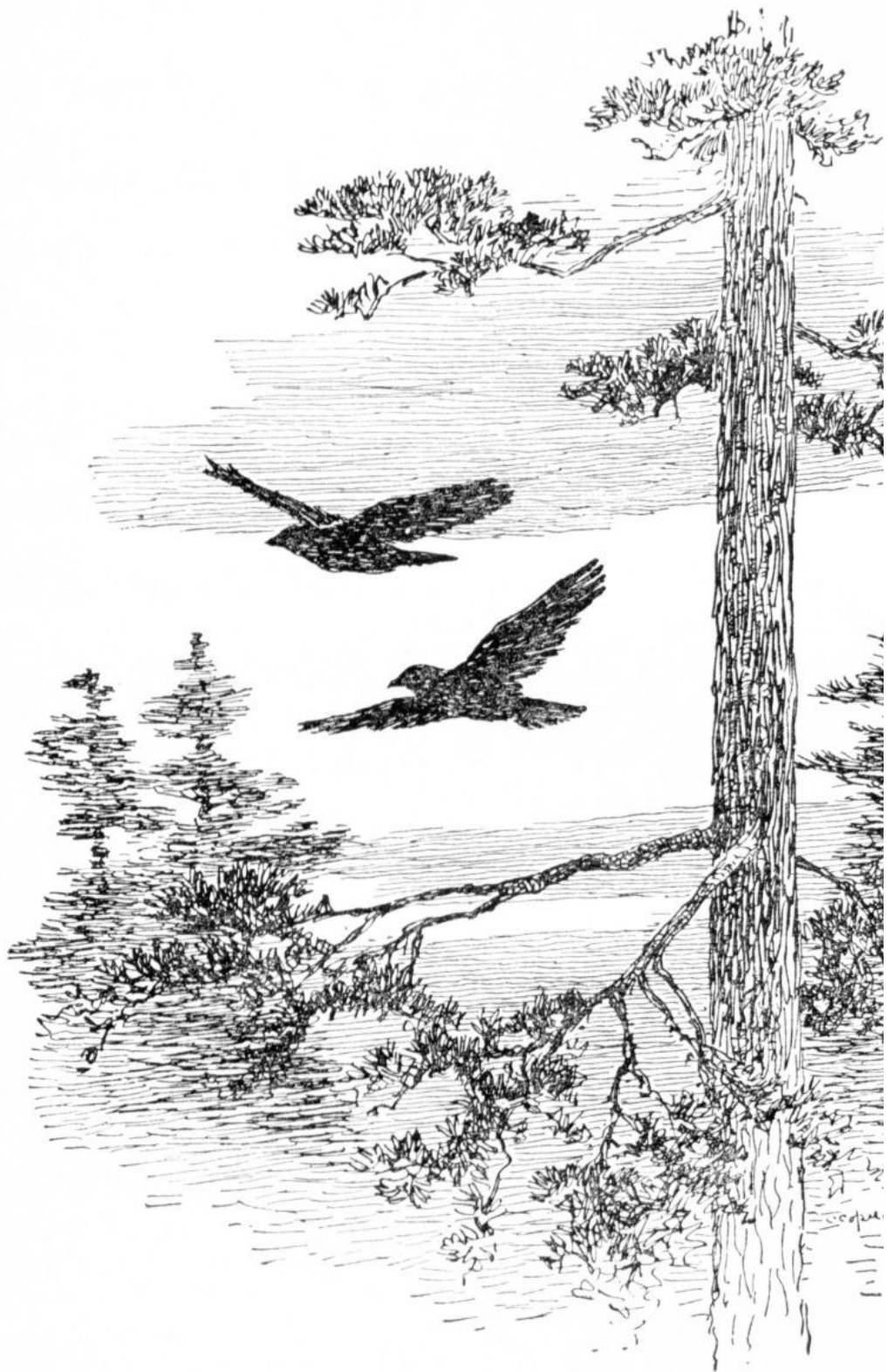
The robin's faint song may have been one of contentment with his lot, or one of evening praise for as many mercies as he had received, but it sounded far more, in that light and that biting air, like the boy who whistles at night on the long and lonesome road to keep his courage up. Then the song died away in his throat, for across the angry crimson of the west flitted silhouetted black wings, and a pair of crows lighted among the thick boughs of the higher pines to roost for the night.

The robin muttered "tut, tut!" somewhat hysterically and slipped away to safer shelter deep among the low boughs and denser shadow of a tree on the edge of the open pasture. No doubt he recognized hereditary enemies of his race, and though he was tough enough to dare a northern winter, was unwilling to take chances with the strong black bills of these reckless freebooters of the wilderness. And he was right. Crows rarely eat grown robins, for they cannot catch them, but the tender, half-fledged nestlings are the mainstay of many a crow saturnalia.

Only too well do I remember an orgy of this sort. It was late May and the scent of the apple blossoms filled all the orchard with delight, just as the robins, morning and evening, filled it with song. They sang for every cloud that crossed the sky and piped up now and then in the full sunshine. How they found time for it all it is hard to tell, for every nest was full of young birds that eat almost their weight in hearty food each day.

One day the tunes changed. Coming into the farthest corner from a woodland trip I heard from some ancient, neglected trees, such as the robins always love and in which

were grouped three or four nests, wild shrieks of anger and dismay from a whole chorus of robins. Coming nearer I could hear crow voices in guttural undertones, croaking ghoulishly.



Across the angry crimson of the west flitted silhouetted black wings

The crow has a language, not exactly of words but of inflections and intonations, which express the primal emotions pretty clearly. I always think I know what he means, though undoubtedly his crow hearers understand the finer shades of inflection better than I do.

There is the shout of warning which says plainly, "Look out, there is trouble right ahead of you!" A similar shout, but with different inflection, says, "Come on. Come on. I'll show you something worth seeing." There is the yell of derision and defiance with which a flock drives an owl through the forest; there is the gentle cooing croak with which mated birds do their love-making. There is the cry of terror and the suppliant call for food from the full-grown young. There is also a peculiarly devilish croak of satisfaction which they make only when feasting on the tender nestlings of pasture birds.

This I knew, and I rushed to the rescue of my young robins, but I was much too late. The feast was well along toward its conclusion and the nests were nearly empty. The parent birds, reinforced by others of the neighborhood, were doing their best. They plunged and darted at the marauders, plucked and clawed at them, but not one whit could they stir them, nor did they leave at my approach, and it took vigorous and well-directed volleys of stones from a near-by heap to drive them away. Then they went heavily, as if gorged to such repletion that they could hardly fly.

I went on home sick at heart, and vowing shot-gun vengeance on all crows thereafter; and it was not until I had carved the chicken for dinner that I realized that there might be extenuating circumstances. For, after all, the crows had as much right to robin for their dinner as I had to chicken for mine.

Crows certainly are responsible for a large amount of infant mortality among young birds in the nesting season, however, and to my mind it is the greatest crime of which these black robbers stand guilty. It is for this reason that the crow is so well hated by smaller birds, and I don't doubt it is this consciousness of guilt that makes him hang his head and flee away before the attack of the least of them. Blackbird and kingbird alike will send him flapping in shamed haste for the big wood, and it makes no difference whether or not he has attempted to burglarize their homes or slaughter their children.

Just as a known pickpocket is railroaded out of town by the police, whether guilty of present misdemeanor or not, so the kingbird sends flying any crow that crosses his path during the nesting season. You will hear the strident, half-hissing scream of rage on the part of the kingbird, see him launch himself from the air above and strike the back of the flapping crow with a thump that perhaps makes the feathers fly. The crow never attempts to strike back. He merely hangs his head and scuttles the faster for the tall timber where is release from this torment. I've never known the kingbird or any other indignant small bird to do the crow material harm; but he certainly sends him flying.

One August, traversing a lonely swamp, I heard a great commotion among crows over in its duskiest, farthest corner. Slipping quietly up, I found a number of them swooping about another, which sat on a low limb within a few feet of the ground. This crow was making beseeching cries, like those of a greedy youngster which still hoped to be fed, and I thought this was the case at first, for, though by August all young crows have long been full grown, the old birds continue to keep oversight of them. I had no sooner come within sight than the keen birds saw me, and away they all went except the supposed youngster, who still kept his perch and his silence, nor did he attempt to move as I approached and finally picked him off his perch.

For he was no youngster, this crow, but was blind, old, and emaciated. I think from the appearance of his eyes that he had been blind for a considerable time, and the interesting question arises as to how he had lived thus far. Surely he could not have found food for himself thus for any long period of time, so perhaps the other crows had fed him right along.

How old crows grow to be I do not know, but whatever extreme age they attain this one was it. I took him home and gave him the freedom of the yard, which he accepted. I fed him, and he seemed to be glad to have a foster parent and to have no fear. But his presence was fiercely resented by another family, and that was the kingbirds that had nested in a neighbor's apple tree. The young were grown up long ago. In fact, the kingbirds had not been seen about for some time, but the crow had no sooner appeared than they came darting into the yard and savagely attacked him.

Again and again I had to rescue him from their fury, though he was the meekest crow I have ever seen, and they no longer had young to defend. Kingbirds go to bed at early dusk as a rule, but even after dark and long after I had put my foundling under shelter for the night, this pair could be heard swearing away to themselves up in the top of their apple tree, waiting for one more whack at him. Kingbirds leave us for the south about the first of September. I am quite sure this pair delayed their migration for some days that year, hating to give up their daily harrying of my ancient and toothless old crone of a crow.

He died, of old age no doubt, before the winter, seeming to fade gently away, as a patriarch should. When, about the fifth of May the next year, the kingbirds came back, they were noticed looking our back yard over very minutely several different times. They remembered the crow and were prepared to drive him over into the next country before they began their nesting.

The patriarch was so old he could not see when I found him. Box and Cox were so young when I lifted them from their nest that they had never seen. They had scarcely kicked their blue-green, brown-splashed eggshells overboard when I climbed to their great, strongly-built home in the upper limbs of a good-sized pine. It had a foundation of stout sticks topped with smaller ones, and within these a well-woven cup of slender twigs lined with grapevine bark and the soft fiber of the red cedar.

There were five young, hideous, negroid creatures with dark warts where eyes would be, and mouths that gaped portentously. Had I realized when I got them the amount of bird food those gaping mouths would engulf, and then opening, clamor for more, I would have left them to their parents. These had slipped silently away when I approached the nest, nor were they visible at all during the kidnapping. I take it that this desertion is prompted by wisdom, not cowardice or heartlessness, for crows are devoted parents and look after their young long after they have left the nest and after a period at which the devotion of other bird parents has ceased.

There was no choice among the five; all were equally ugly, and I took two at random and shinned down the tree with them in a bandanna handkerchief swung from my teeth. Seeing their young thus carried away in the teeth of a marauder, I dare say the old crows thought of me as I thought of their fellows that ate the young robins. But though I don't doubt they saw from safe retreat all that went on, they took great care neither to be seen nor heard.

The two young birds accepted the featherless biped in *loco parentis* without any question. They also accepted all I would put into their yawning maws, and opened them mutely for more. By and by they found eyesight, and later voices. Then, not seeing food coming, they would call for it with yearning and yell for it with ebullient eagerness when they saw it, or me, or any other approaching biped. I don't think the neighbors took kindly to this pair of pets of mine. It was too much like having a piano and an opera candidate in the next flat.

Sometimes their own weight a day went into these howling dervishes, in the form of fish, frogs, grasshoppers, meat, scraps from the table, any thing, indeed, that luck put in my way or that the ingenuity of desperation suggested, and still nightfall found them ravenously emulating Oliver Twist. But they grew, and grew so much alike that which was Box or which was Cox neither I nor anybody else could tell.

As their feathers sprouted so did their ambitions. In a little while they could stand on the edge of their nest, which I had built for them in the low limbs of a tree near the back door, and flap their impotent wings at the same time that they yelled for the waiter. Though I was their guardian angel it was not for me in particular that their clamor rent the sky, but any one who by any remote possibility might feed them.

Their first venture off the nest showed this. The new minister went through the yard, thus making a short cut to a neighborly call. By chance Box and Cox had been stuffed to repletion some minutes before and were silent, half asleep in fact. But when the new minister's hat passed within two feet of their nest they rose to the occasion, and with one mutual crow-language yell of "Bread, for the Lord's sake give us bread!" they landed on his hat. The family rescued him, of course, with humble apologies, and he was good enough not to take offence. He came later to call, generously, also I think somewhat stealthily, and by way of the front door.

Box and Cox had found their wings and they used them to hunt down all possible purveyors of food. They knew me best because I fed them oftenest, but otherwise

showed neither partiality nor affection. They kept away from the carpenters at work in the near-by shop because they had many times narrowly missed decapitation with hatchets, but they kept just beyond hatchet stroke only and clamored tantalizingly. The carpenters thought they taunted them and used to threaten gun play.

In return the crows stole bright nails, screws, and such small tools as they could get hold of. They got away with my pearl-handled pocketknife on the same principle, and though we often hunted for their hoard we never found it. Their doings were often amusing to the bystander, but more often vexatious and sometimes outrageous. I have still a vivid mental picture of good old Grandfather Totter on his way home by the path in the field, and stalled, because he could no longer use his cane to hobble with, but had to have it to fight off Box and Cox.

Bird neighbors did not love Box and Cox any better than did human neighbors, and their presence kept kingbirds and robins, bluebirds and sparrows all in a state of great nervous tension, though I am bound to say that I never knew the crows to disturb their nests or young. In fact, as long as I had them, Box and Cox showed no signs of learning to forage for themselves in any way. They depended absolutely on mankind for food, and if man was not kind they went hungry. I think that if I had conscientiously tried to wean them they would have shown ability to take care of themselves, but I never had the courage to try. I did not think the neighborhood would stand the racket.

One day they simply disappeared and I never knew what became of them. Perhaps they suddenly heard and answered the call of the wild. The neighbors had been wild more than once.

Box and Cox were a disappointment. They showed little of either wit or wisdom. They had a small amount of roguishness and a mighty appetite. Such traits as they showed were those of youth; those they lacked might have come with age. Perhaps parent crows teach their young the wisdom which wood-bred birds certainly show. Box and Cox had none of it, or if they had they hid it with the pocketknife and the carpenter's tools.

On the other hand, the strongest trait of the wood-bred crow is his distrust of man. Instinct, if it works in the crow tribe, should certainly have implanted this distrust in the youthful heads of Box and Cox, but they showed nothing of the sort. And there you have the crow puzzle all over again, for the crow, wild or tame, is a puzzle. Half a hundred of them the other day were congregated about a wood road through the pines, yelling themselves hoarse in the wildest of excitement.

So interested were they that they took no notice of me when I approached, thinking that they had a hawk or owl at bay there and were harrying him. So I walked down the wood road right in amongst them. But there was neither hawk nor owl nor anything else there to account for their excitement. They tore about this empty space, cawing, fluttering, standing erect, alert, and quivering on a limb and gazing wildly at what seemed to be to them very real and very terrible. But it was nothing to me; I

could not find so much as a chipmunk stirring there. After a little they chased this terrible nothing on down the road and then across lots into another part of the wood, leaving me gaping and in doubt whether they were just playing a game among themselves, all making believe they saw a monster where there was none, or whether they really could see some woodland bogle that was invisible to my dull eyes and were following him on his way.

Box and Cox may have been among them, and for all I know may later have told the crowd what a queer creature man is when you come to know him as foster-fathered crows have to.

INDEX

A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, V, W.

A

Acorns, 98, 99, 100
Admiral, red, 5
Alder, 177, 182
—— red-berried, 195
Angleworms, 160
Anosia plexippus, 117
Antiopa vanessa, 116
Apple blossoms, 119, 198
—— tree, 27, 135, 205
—— wild, 156
Arbor vitæ, 39
Aroostook war, 24
Ash tree, 96, 133
Aster, 114
Azalia, 166

B

Barberry, 184, 185
Bat, 76, 79, 82
Bayberry, 111
Bee, 10
Beech, 29
Birch, 29, 34, 35, 74, 75, 83, 92, 93, 126, 127, 133, 143, 165
—— C. T. U., 92, 94
Bittern, 60
Blackberry, 16
Blackbird, red-winged, 119, 178, 180, 182, 201
Blueberry, swamp, 164
Bluebird, 113, 212
Blue Hills, 110
Buck, 38
Buckthorn, 195
Butterfly, 5, 114, 117
—— admiral, red, 5
—— Anosia plexippus, 117
—— Antiopa vanessa, 116

—— Hunters', [115](#)
—— monarch, [114](#), [116](#), [117](#)
—— painted lady, [5](#), [115](#)
—— Pyrameis, [117](#)
—— Pyrameis atalanta, [5](#)
—— Pyrameis cardui, [5](#), [116](#)
—— sulphur, [114](#)

C

Catbird, [181](#), [182](#), [183](#)
Cedar, [143](#)
—— berries, [113](#)
—— pasture, [121](#)
—— red, [113](#), [206](#)
—— white, swamp, [113](#)
Cemetery Hill, [173](#)
Cherries, [16](#)
Chestnut, [67](#), [68](#), [69](#), [71](#), [75](#), [76](#), [79](#), [83](#), [95](#), [96](#), [98](#), [99](#), [109](#), [155](#)
—— bur, [77](#)
—— leaves, [83](#)
—— tree, [76](#), [77](#)
Chickadee, [3](#), [4](#), [173](#)
Chipmunk, [214](#)
Christmas, [195](#)
—— tree, [35](#)
Clam, [41](#)
Clintonia borealis, [15](#), [16](#), [17](#)
Clover, [6](#)
Cocoanut, [19](#)
Coon, [174](#)
Cowbird, [188](#), [189](#), [190](#), [191](#)
Coyote, [38](#)
Crow, [40](#), [197](#), [198](#), [199](#), [201](#), [202](#), [203](#), [204](#), [205](#), [207](#), [211](#), [212](#), [213](#), [214](#), [215](#)
—— nest, [143](#)
Crustaceæ, [51](#)
Currant, [17](#)
—— fairy, [18](#)
Cypripedium acaule, [35](#)

D

Deer, [27](#), [37](#), [38](#)
Dendragapus canadensis, [35](#)

Doe, [38](#)
Duck, [52](#), [53](#), [62](#)
—— black, [54](#)
—— “spirit,” [57](#)
—— teal, blue-winged, [48](#), [49](#)

E

Elder, [177](#)
Elm, [133](#)
Epilobium angustifolium, [11](#)
Erechthites, [11](#)
—— hieracifolium, [9](#)

F

Fawn, [38](#)
Fern, cinnamon, [184](#)
—— wood, [88](#)
Fir, [23](#), [29](#), [32](#), [33](#), [34](#), [35](#), [37](#)
Fireweed, [10](#), [11](#), [12](#)
Fish, flying, [110](#)
Flicker, [40](#)
Fox, [38](#), [157](#), [174](#)
Frog, [174](#), [208](#)

G

Glow-worm, [20](#)
Goldenrod, [114](#)
Goliaths, [28](#)
Grape, [74](#), [114](#)
Grapevine, wild, [187](#), [206](#)
Grass, purple wood, [73](#), [82](#)
Greece, [27](#)
Grebe, pied-billed, [52](#), [53](#), [54](#), [55](#), [56](#), [57](#), [62](#)
Greek, [191](#)
Grouse, Canada, [3](#)
—— ruffed, [35](#), [37](#)
Gulliver, [28](#)

H

Hackmatack, [39](#)
Hawk, [38](#), [214](#)
Hazel nuts, [155](#)

Hedgehog, [157](#)
“Hell-diver,” [55](#)
Hemlock, [121](#)
Hickory, [79](#), [80](#), [82](#), [88](#), [109](#), [153](#), [154](#), [155](#)
Hob, [8](#)
Holly berries, [195](#)
Huckleberry, black, [164](#)
Hunters’ butterfly, [115](#)

I

Ignis fatuus, [90](#)
Indian, [30](#)
—— summer, [152](#)

J

Jay, blue, [111](#)
Joepy weed, [177](#)
June berries, [156](#)
Juniper, [125](#), [126](#)

K

Katahdin, [23](#)
Kimball, George, [33](#)
Kingbird, [201](#), [202](#), [204](#), [205](#), [212](#)

L

Lady’s slipper, [15](#), [16](#)
Leprachauns, [13](#)
Lilac, [82](#)
—— purple, [134](#)
—— white, [134](#)
Liliputians, [28](#)
Locusts, [103](#)
Loon, [62](#)

M

Macwahoc-Kingman road, [29](#)
Maple, [34](#), [89](#), [91](#), [133](#), [136](#), [137](#)
—— red, [94](#), [96](#)
—— Norway, [94](#), [97](#), [135](#)
—— silver-leaved, [94](#)
—— swamp, [89](#), [90](#), [93](#), [94](#)

—— white, 94
“Mast,” 100, 101
Milkweed, 177, 178, 186
Mistletoe, 195
Mitchella, 17
Monarch, 114, 116, 117
Mouse, field, 183

N

Norse Sagas, 46
Nuthatch, 176, 177

O

Oak, 97, 98, 101, 102, 103, 133, 181, 182
—— black, 98, 99, 101, 103
—— black, “mast,” 101
—— red, 103
—— scarlet, 103
—— scrub, 103, 110
—— white, 98, 99, 100, 101, 103
Oak, white, “mast,” 101
Oliver Twist, 209
Orchid, 15
Owl, 214
—— barred, 40
—— screech, 174

P

Painted lady, 5, 115
Palm, 19
Partridge, 14, 157, 158, 162
—— berries, 14, 16, 17
—— birch, 35, 36
—— spruce, 35, 36
Patten Road, 23, 24, 27
Pear tree, 35
Petrel, 47
Pine, 15, 32, 114, 174, 197, 206, 214
—— pitch, 5, 121, 122, 123, 125, 126, 127
—— pumpkin, 29
—— white, 120, 125
“Piney Home,” 33

Plover, [62](#)
—— piping, [51](#)
—— ring-necked, [51](#)
—— yellow-leg, [47](#), [48](#), [49](#)
Poa serotina, [58](#)
Pokeberry, [13](#)
Pokeweed, [12](#)
Porcupine, [27](#), [157](#)
Porzana carolina, [59](#)
Proteus, [9](#)
Pyrameis, [117](#)
—— atalanta, [5](#)
—— cardui, [5](#), [116](#)

Q

Queen Mab, [17](#)

R

Rabbit, jack, [38](#)
Rail, Carolina, [57](#), [58](#), [59](#), [60](#), [61](#), [62](#)
Raspberry, [16](#)
Rivers, Mattawamkeag, [29](#)
—— Moluncus, [29](#), [30](#), [33](#)
—— Macwahoc, [29](#), [33](#)
—— Orinoco, [58](#)
—— Amazon, [58](#)
Robin, [111](#), [112](#), [113](#), [195](#), [196](#), [197](#), [198](#), [200](#), [201](#), [212](#)

S

Sandpiper, spotted, [49](#), [51](#)
Sage-brush, [38](#)
“Seasons,” by Thomson, [136](#)
Shadbush, [156](#)
Skunk, [157](#)
Smilacina bifolia, [17](#)
South African mines, [74](#)
Sparrows, [212](#)
—— English, [190](#), [191](#)
Spruce, [32](#), [34](#), [35](#), [36](#), [37](#), [39](#)
—— cat, [29](#), [33](#)
—— black, [33](#)
—— timber, [33](#)
—— white, [33](#)

Squirrel, [69](#), [70](#), [72](#), [73](#), [144](#), [152](#), [154](#), [155](#)
—— gray, [70](#), [72](#), [79](#), [80](#), [81](#), [82](#)
—— red, [67](#), [68](#), [70](#), [72](#), [143](#), [144](#), [183](#)
Sulphur butterfly, [114](#)
Sumac, [82](#)

T

Teal, blue-winged, [48](#), [49](#), [54](#)
Telia polyphemus, [78](#), [83](#)
Thoreau, [26](#)
Toad, tree, [174](#)
Totter, Grandfather, [211](#)
Trillium, [19](#), [20](#)
Triton, [9](#)

V

Vikings, [47](#)
Vireo, [187](#)
Virgin's bower, [177](#)

W

Warbler, [112](#), [188](#), [189](#)
—— myrtle, [112](#)
—— yellow, [184](#), [185](#), [186](#), [187](#), [190](#), [191](#)
Willow, [97](#)
—— herb, [11](#)
Witch hazel, [114](#), [118](#)
—— blooms, [120](#)
—— nuts, [118](#)
Woodchuck, [6](#), [7](#), [8](#)
Woodcock, [159](#), [160](#), [161](#), [162](#)
Wood mice, [18](#)
Woodpecker, golden-winged, [39](#), [40](#)
—— partridge, [114](#)
Wordsworth, [9](#), [136](#)
Wrights, [41](#)

*** END OF THE PROJECT GUTENBERG EBOOK WOOD
WANDERINGS ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.

WOOD WANDERINGS



WINTHROP PACKARD

[back](#)



[back](#)



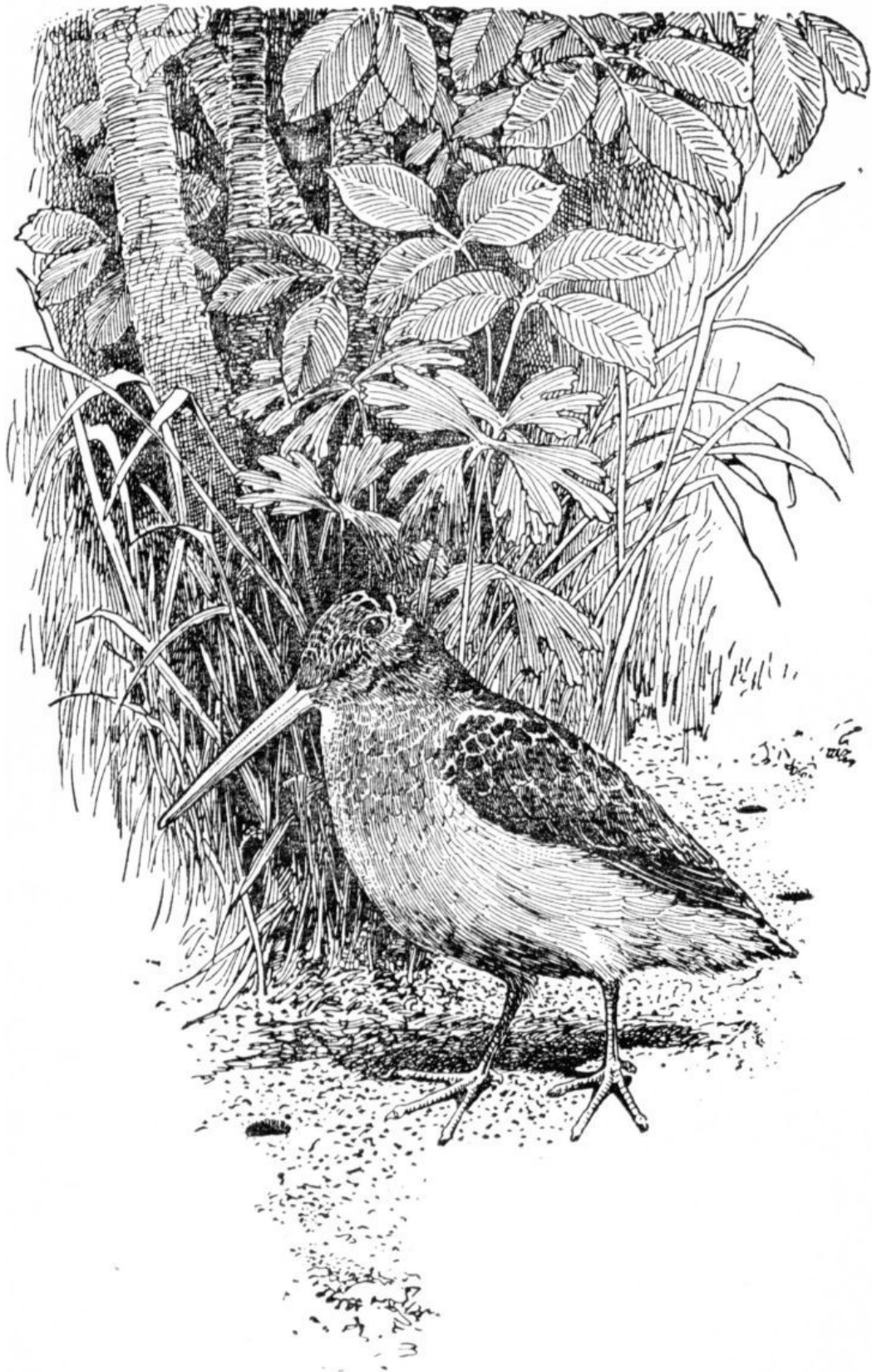
[back](#)



[back](#)



[back](#)



[back](#)



[back](#)



[back](#)